Weakly Supervised Hyperspectral Target Detection Based on Contrastive Learning and Graph-Structural Position

Ying Cui, Xiaoqi Ren, Liguo Wang, *Member, IEEE*, Haifeng Zhu, Zhongwei Shen, *Member, IEEE*, and Chunhui Zhao

***Abstract*—Hyperspectral target detection is significant in remote sensing, yet existing methods heavily rely on strong supervision. Target prior spectra are sensitive to illumination, atmospheric, and sensor variations, causing inevitable deviations that degrade performance. Pixel-level annotations are prohibitively costly due to high spatial resolution and subpixel localization difficulties. Multiple instance learning enables weakly supervised detection using region-level labels, but most existing methods neglect spatial dependencies among pixels and suffer from limited localization capability. To address these issues, this article proposes a graph-structured positional transformer for multiple-instance learning (GSPT-MIL), a graph-structured and position-aware weakly supervised multiple-instance learning (MIL) method that achieves accurate pixel-level detection using only bag-level labels, without requiring target priors or pixel-level annotations. The method performs iterative optimization under an expectation-maximization (EM) framework. The expectation step (E-step) infers target locations via a local similarity graph convolution module (LS-GCM) that models spatial dependencies, while the maximization step (M-step) fuses absolute position information via a multiscale contextual position encoder and enhances feature discriminability via a spectral–spatial feature alignment network (SS-FAN). Extensive experiments demonstrate that GSPT-MIL consistently outperforms existing weakly supervised and conventional detection methods, validating its effectiveness and superiority.**

***Index Terms*—Expectation-maximization (EM) algorithm, graph convolutional network, hyperspectral target detection, multi-instance learning, position encoding.**

Received 1 June 2026; revised 5 September 2026; accepted 14 September 2026. Date of publication 19 September 2026; date of current version 2 October 2026. This work was supported in part by the New Round of "Double First Class" Discipline Collaborative Innovation Achievement Project in Heilongjiang Province under Grant LJGXCG2024-F03, in part by the National Natural Science Foundation of China under Grant 62671109 and Grant 62371153, in part by China Postdoctoral Science Foundation under Grant 2025M784399, in part by the Fundamental Research Funds for the Central Universities under Grant 3072025CFJ0807, in part by Heilongjiang Postdoctoral Fund under Grant LBH-Z25121, in part by the Major Scientific and Technological Achievement Industrialization Project of Heilongjiang Province under Grant CG24018, and in part by the Open Fund of the National Key Laboratory of Radar Detection and Sensing under Grant KGJ24010742402. *(Corresponding author: Haifeng Zhu.)*

Ying Cui, Xiaoqi Ren, Haifeng Zhu, Zhongwei Shen, and Chunhui Zhao are with the College of Information and Communication Engineering, the Key Laboratory of Advanced Marine Communication and Information Technology, Ministry of Industry and Information Technology, and Heilongjiang Collaborative Innovation Center of Advanced Intelligent Perception Technology, Harbin Engineering University, Harbin 150001, China (e-mail: cuiying@hrbeu.edu.cn; renxiaoqi@hrbeu.edu.cn; zhuhaifeng@hrbeu.edu.cn; shenzhongwei@hrbeu.edu.cn; zhaochunhui@hrbeu.edu.cn).

Liguo Wang is with the College of Information and Communication Engineering, Dalian Minzu University, Dalian 116600, China (e-mail: wangliguo@hrbeu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2026.3734980

I. Introduction

HYPERSPECTRAL imaging can acquire spectral information in hundreds of narrow bands [1], enabling accurate identification and discrimination of ground objects based on their unique spectral signatures [2]. The intrinsic integration of spectral and spatial information makes it possible to achieve precise target detection at the pixel or even subpixel level [3]. In recent years, pixel-level and subpixel target detection has been widely applied across various domains, demonstrating significant value in fields such as land cover classification [4], [5], maritime and land-based military target detection [6], urban planning [7], and crop disease detection [8].

The detection of hyperspectral targets mainly aims at identifying targets within complex backgrounds by exploiting subtle spectral differences between the target and the background in hyperspectral imagery [9]. Depending on whether prior knowledge of the target is used, hyperspectral target detection can be categorized into anomaly detection and traditional target detection. Anomaly detection does not require a prior target and focuses on identifying anomalies from complex backgrounds; however, its performance critically depends on the accuracy of background modeling [10]. In contrast, traditional target detection relies on target spectral priors and is highly sensitive to spectral fidelity. In practical scenarios, factors such as illumination variation, atmospheric scattering, sensor noise, and spectral variability of ground objects can substantially degrade detection performance. Both types of methods face inherent technical bottlenecks. The former depends on background modeling precision and tends to produce numerous false positives in complex backgrounds or when the spectral difference between target and background is small. The latter relies on accurate target spectra and exhibits fragile performance under real-world deviations [11]. More critically, both approaches depend on strong supervision signals, such as precise pixel-level annotations or target spectral information. Given the high spatial resolution of hyperspectral images, subpixel targets are difficult to accurately identify visually. Manual pixel-level labeling is time-consuming, labor-intensive, and prone to subjective variability, severely restricting the large-scale application of conventional methods in real-world scenarios.

To overcome the bottleneck of strong supervision dependence, multiple-instance learning (MIL) has been introduced into hyperspectral target detection, offering a weakly supervised solution [12]. MIL partitions the training data into bags

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:51:51 UTC from IEEE Xplore. Restrictions apply.

and requires only bag labels instead of instance-level labels, where a positive bag contains at least one target instance, and a negative bag consists entirely of background instances [13]. In hyperspectral target detection, annotators can select regions containing targets as positive bags based on visual inspection or GPS coordinates, without the need for precise pixel-level location annotations. This significantly reduces annotation costs [14] and enables a critical transition from strong supervision to weak supervision.

Early MIL-based hyperspectral target detection methods focused primarily on target feature learning and can be categorized into two types. The first type comprises target feature representation-based methods, which learn spectral feature prototypes of targets from weakly annotated bags and subsequently combine them with conventional detection models to achieve target localization. For example, the extended functional multiple-instances (eFUMIs) method [15], [16] is based on a linear spectral mixture model and learns a single target spectrum alongside a background dictionary by iteratively maximizing posterior probability. However, this approach aggregates all positive bags into a single entity, neglecting the unique information contained within individual bags. Methods such as the multiple-instance adaptive coherence estimator (MI-ACE), the multiple-instance spectral match filter (MI-SMF) [17], and their extensions [18], [19] learn discriminative target features by maximizing the detection response for each positive bag while minimizing that for negative bags. Nevertheless, these methods can only learn a single target prototype and struggle to address spectral variability. To overcome this limitation, approaches such as the multitarget hybrid estimator (MI-HE) [20], [21], the multiple-instance multiple-diversity adaptive coherence estimator (MIMD-ACE) [22], and the multitarget multi-instance adaptive cosine estimator (MTMI-ACE) [23] have been extended to multitarget feature learning, accommodating spectral variability by learning multiple target prototypes. However, these methods still rely on hand-crafted statistical assumptions and shallow spectral features, limiting their capacity to capture deep nonlinear characteristics in complex scenarios [24]. The second type comprises instance selection-based methods. For instance, the MIL via embedded instance selection (MILES) [25] maps bags into a high-dimensional feature space defined by representative instances and subsequently performs feature extraction and classification using a 1-norm support vector machine (SVM). However, this method is highly sensitive to instance selection and may introduce redundant information when the background is complex.

With the advancement of deep learning, multiple instance learning methods based on deep neural networks have achieved breakthrough progress in hyperspectral target detection, leveraging their powerful deep feature extraction capabilities. The core challenge of such methods lies in how to identify true target instances from positive bags while avoiding false-positive interference. The attention mechanism, in particular, enables the model to adaptively evaluate the importance of each instance, thereby achieving focused feature aggregation and fusion [26]. Early deep MIL methods typically employed predefined permutation-invariant pooling for bag-level feature fusion. However, such pooling functions are fixed and non-trainable, making it difficult to adapt to varying scenarios. To address this, Wang et al. [27] proposed the max-attention MIL network (MINN), which trains a detector using only the instance with the highest attention weight in a bag, but this approach neglects most of the other instances, leading to limited performance. Building on this, Jiao et al. [28] introduced L-regularized attention MINN (L1-attention MINN) by imposing an L1 sparsity constraint on attention weights, forcing false-positive instances to receive near-zero attention and thereby enhancing the discriminability of target features. Meanwhile, Ilse et al. [29] proposed attention-driven deep multiple instance learning (Attention-DMIL), where trainable attention weights adaptively assign the contribution of each instance to the bag label; this has become a core framework. Nevertheless, existing deep MIL methods still suffer from insufficient modeling of interinstance correlations. Most methods are based on the independent and identically distributed (i.i.d.) assumption, ignoring the spatial continuity and spectral correlations among neighboring pixels. In real scenes, ground object distributions often exhibit local clustering and spatial structure, and such non-i.i.d. contextual information is crucial for improving model discriminability [30]. Moreover, existing methods rely almost exclusively on spectral features and fail to effectively exploit the absolute position information of each pixel in the global image coordinate system. In urban areas or complex terrains, objects with similar spectra but distinct spatial locations are very common, making them difficult to distinguish using spectra alone [31]. Furthermore, as heterogeneous modalities, spectral and spatial positional information have not yet been effectively co-represented and aligned within existing MIL frameworks, resulting in insufficient feature discriminability and limited generalization capability. To address the first issue, Yang et al. [32] proposed variational multiple instance learning embedded correlation modeling (VMIL-ECM), which is based on the non-i.i.d. assumption. This method captures interinstance correlations by stacking Transformers and optimizes the detection model using the expectation-maximization (EM) algorithm along with a dynamic threshold strategy, thereby improving adaptability to complex scenarios. However, this method still does not resolve how to effectively incorporate absolute positional information and achieve deep fusion of spectral–spatial multimodal features. The self-attention mechanism it relies on primarily captures relative relationships among instances rather than global absolute coordinates, leading to insufficient localization accuracy and spatial consistency constraints when applied to large-scale hyperspectral scenes with complex spatial layouts.

To address the above issues, this study proposes a graph-structured and position-encoded transformer-based multiple instance learning model, termed graph-structured positional transformer for MIL (GSPT-MIL). Under the EM framework, the model decouples target localization and detector training into a differentiable graph convolution module and a position-aware module, forming a closed-loop iterative optimization system. For spatial relationship modeling, a local graph message passing mechanism is designed to generate high-quality pseudo labels from weak labels by exploiting spatial

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:51:51 UTC from IEEE Xplore. Restrictions apply.

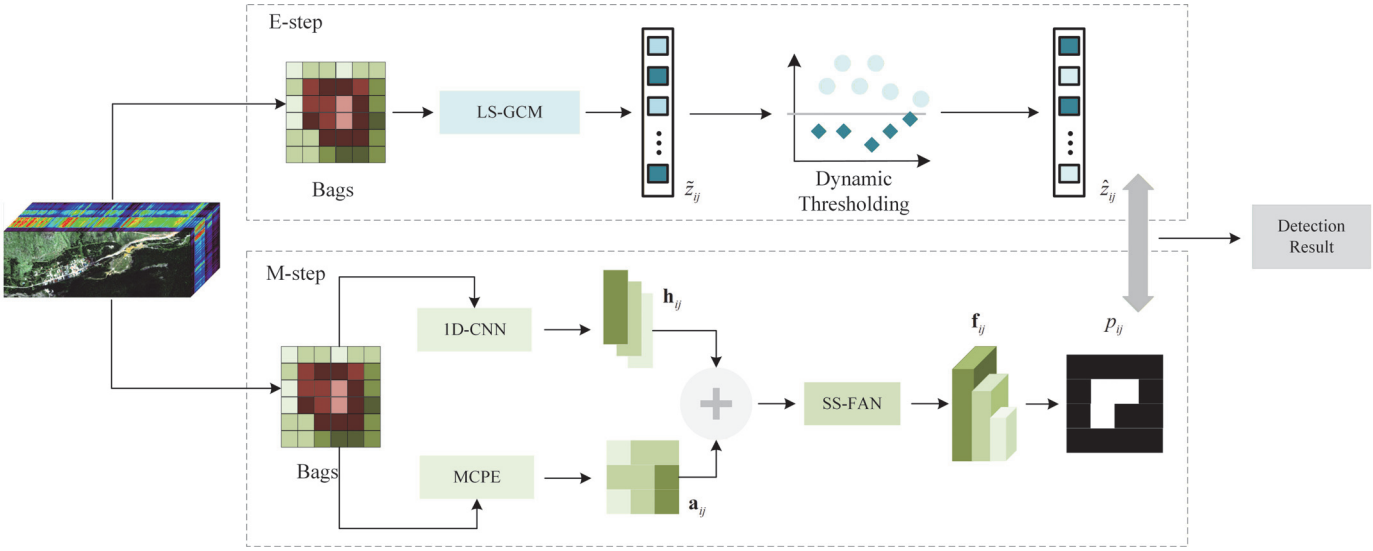


Fig. 1. Overall framework diagram of the GSPT-MIL model.

consistency among neighboring pixels. For positional information exploitation, a multiscale adaptive position encoding method is proposed, which generates sparse positional features through tunable grid partitioning and sliding window statistics, thereby enhancing the representation capability for complex spatial layouts. For feature fusion, a regression-based cross-view contrastive learning approach is introduced, which uses the continuous similarity between spectral features and positional features as a self-supervised signal to promote cross-modal feature alignment. Specifically, the main improvements and innovations of this work are as follows.

1) To address the problem of entangled optimization objectives and low training efficiency in end-to-end models for weakly supervised hyperspectral target detection, this article proposes a weakly supervised learning framework that synergizes graph structures with EM, termed GSPT-MIL. The framework decouples target localization [expectation step (E-step)] and detector training [maximization step (M-step)] into a graph convolution module and a position-aware module, respectively, forming a closed-loop optimization system with efficient synergy, thereby effectively improving the efficiency and stability of weakly supervised training.
2) To tackle the challenge of modeling complex spatial distributions of targets and the interdependencies among pixels, this article designs a local similarity graph convolution module (LS-GCM). By explicitly constructing pixel graphs and aggregating neighborhood information, this module directly learns target distributions that conform to spatial consistency from weak labels, achieving accurate pixel-wise localization and enhancing the modeling capability for complex spatial structures.
3) To address the insufficient fusion of heterogeneous spectral and spatial features, this article collaboratively constructs a multiscale contextual position encoder (MCPE) and a spectral–spatial feature alignment network (SS-FAN). The MCPE generates absolute position encodings to preserve spatial layout information, while the SS-FAN aligns spectral features with position encodings in a latent space through contrastive learning. Their synergistic learning produces a more discriminative unified feature representation, significantly improving the quality of feature fusion and the robustness of detection.

II. Proposed Method

A. Overall Framework of GSPT-MIL

The proposed GSPT-MIL method constructs an iterative optimization framework based on the EM algorithm to address the problem of hyperspectral target detection using only bag-level labels. This framework strictly adheres to the mathematical principles of the classical EM algorithm, progressively optimizing model parameters and inferring pixel-level target locations by alternately executing the E-step and M-step) Fig. 1 illustrates the overall architecture and data flow of the GSPT-MIL model.

The figure depicts the iterative workflow of GSPT-MIL based on the EM algorithm. On the left are the input hyperspectral image patches and their bag-level labels. The upper part of the workflow represents the E-step, where spectral data is processed by the LS-GCM module to generate pixel-level pseudo-labels $\tilde{z}_{ij}$, which are then refined via a dynamic threshold to produce $\hat{z}_{ij}$. The lower part represents the M-step: spectral features $\mathbf{h}_{ij}$ are extracted by a 1D-CNN and fused with position encodings $\mathbf{a}_{ij}$, generated by the MCPE, to obtain $\mathbf{f}_{ij}$; the SS-FAN module applies contrastive learning constraints to the features before and after fusion. The fused features are then fed into a classifier to predict the probability of a pixel being the target. The outputs of the E-step and M-step are jointly used to compute the loss and update parameters. The arrows indicate the direction of iterative optimization.

First, we provide an overview of the EM optimization framework. Given the dataset defined as follows: spectral data $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_{N_t}] \in \mathbb{R}^{d \times N_t}$ and bag-level labels $\mathcal{L} = [L_1, L_2, \ldots, L_K] \in \{0, 1\}^K$. The objective of the model is to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:51:51 UTC from IEEE Xplore. Restrictions apply.

learn a set of parameters Θ that maximize the log-likelihood of the observed labels given the data, denoted as $\log p(\mathcal{L} \mid \mathbf{X}; \Theta)$. Since the true pixel-level labels l_{ij} are unobservable, their relationship is implicitly constrained within the bag-level labels, as shown in (1). We introduce a latent variable $z_{ij} \in \{0, 1\}$ to explicitly model the potential target state of the ith pixel in the jth bag

$$L_i = \begin{cases} 1, & \text{if } \exists j,\ l_{ij} = 1 \\ 0, & \text{if } \forall j,\ l_{ij} = 0. \end{cases} \tag{1}$$

The EM algorithm solves this optimization problem by iterating the following two steps.

1) *E-Step:* Given the current fixed parameters $\Theta^{(r)}$, compute the posterior expectation of the latent variable $z = z_{ij}$ to construct the expectation function (Q-function) of the complete-data log-likelihood.
2) *M-Step:* Update the model parameters $\Theta^{(r+1)}$ by maximizing this Q-function.

Then, we introduce the workflow of this framework. Different from traditional approximation methods based on variational inference, GSPT-MIL employs neural networks to directly parameterize the posterior estimation of the E-step and the likelihood model of the M-step. This achieves powerful function fitting capabilities while maintaining the theoretical rigor of the algorithm. Each EM iteration of GSPT-MIL comprises the following core processes.

1) E-Step (Target Location Inference): In the E-step, the goal of the model is to estimate the posterior probability of each pixel being a target, given the current detector state. This step is implemented by a target location inference network. The network takes as input the spectral features of all pixels within a data bag $\mathbf{B}_i$ and explicitly models the spatial dependencies between pixels.

Specifically, we construct a graph structure with pixels as nodes using the LS-GCM. Graph neural networks are utilized to aggregate neighborhood information, and the network outputs pseudo-labels $\tilde{z}_{ij}^{(t)}$ for each pixel

$$\tilde{z}_{ij}^{(t)} = q_\phi \left(z_{ij} = 1 \mid \mathbf{B}_i \right) \in [0, 1] \tag{2}$$

where ϕ represents the parameters of the E-step network, i.e., all learnable weights and biases in the graph convolution layers and the subsequent multilayer perceptron (MLP) layers. This output can be interpreted as the confidence that pixel $\mathbf{x}_{ij}$ is a target, based on the current model's understanding. To generate a more reliable supervisory signal, we introduce a dynamic thresholding strategy to binarize the pseudo-labels within each positive bag, obtaining the pseudo-target location estimate $\hat{z}_{ij}^{(t)}$ for supervising the subsequent M-step.

2) M-Step (Detector Parameter Learning): In the M-step, the model utilizes the pseudo-labels $\hat{z}_{ij}^{(t)}$ generated by the E-step as supervisory information to update and optimize the target detector. The detector learning process fully integrates spectral and spatial information.

For spectral feature extraction, we employ a 1-D convolutional neural network (1D-CNN) to extract deep feature representations $\mathbf{h}_{ij}$ from the raw spectrum, where $\mathbf{h}_{ij} \in \mathbb{R}^L$ denotes the feature dimension. Concurrently, we generate an absolute position sequence (APS) $\mathbf{a}_{ij} \in \mathbb{R}^{N_p}$ for each pixel via the MCPE, where N_p represents the sequence length. This sequence is projected through a linear layer and then fused with the spectral features

$$\mathbf{f}_{ij} = \mathbf{h}_{ij} + \mathbf{W}_a \mathbf{a}_{ij} \tag{3}$$

where $\mathbf{W}_a \in \mathbb{R}^{L \times N_p}$ is a learnable weight matrix. This additive fusion ensures that the model perceives both spectral characteristics and absolute spatial positions simultaneously.

The fused features $\mathbf{f}_{ij}$ are then fed into a classifier to predict the probability p_{ij} of a pixel being a target. Simultaneously, to align the representation spaces of the spectral features $\mathbf{h}_{ij}$ and the spatial features $\mathbf{a}_{ij}$ encoded by the position sequence, we introduce the SS-FAN and construct a contrastive learning loss term. This enhances the model's feature discriminability under weak supervision.

3) Iterative Optimization and Convergence: The aforementioned E-step and M-step are performed alternately, forming a complete EM iteration cycle. In the initial iterations, a warm-up phase is established, using simple heuristic rules to initialize pseudo-labels for training stabilization. As iterations progress, the location estimates generated by the E-step become increasingly accurate, thereby providing higher-quality supervisory signals for the M-step. Conversely, the detector optimized in the M-step enhances the reliability of the location inference in the E-step. This co-evolution mechanism enables the model to progressively approximate the true distribution of target pixels until convergence.

B. Local Similarity Graph Convolution Module

To address the challenges of ambiguous target localization under weak supervision and the neglect of spatial correlations between pixels, we designed the LS-GCM. Existing methods often assume that instances within a bag are independent and identically distributed. However, in hyperspectral imagery, adjacent pixels typically exhibit similar material properties. To this end, LS-GCM explicitly constructs a pixel graph for each data bag, where nodes represent pixels and edges are defined jointly by spectral similarity and spatial proximity. Graph convolution is then utilized to aggregate neighborhood information, generating smoother and more consistent target probability estimates. As shown in Fig. 2, this module replaces traditional parameter-free aggregation operations (such as max or mean pooling) with a learnable message-passing mechanism. This enables the model to adaptively learn local contextual information, thereby producing more reliable pseudo-labels during the E-step and effectively improving the training quality of the subsequent detector.

The core idea of LS-GCM is to treat pixels in a hyperspectral image as graph nodes and establish edge connections based on the spectral similarity and spatial proximity between pixels. For an input data bag, we construct an undirected graph $G_i = (V_i, E_i)$, where the node set V_i corresponds to all pixels within the bag, and the edge set E_i represents the connections between pixels.

Unlike methods that rely on fully-connected self-attention or simple pooling, LS-GCM explicitly constructs a local graph under the dual constraints of spectral similarity and spatial

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:51:51 UTC from IEEE Xplore. Restrictions apply.

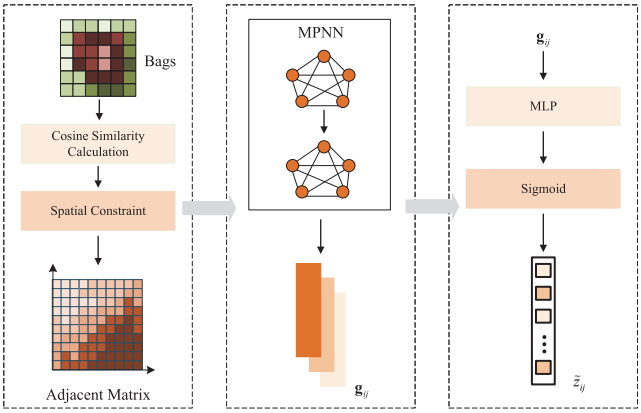


Fig. 2. Architecture diagram of the LS-GCM.

proximity, performing message passing through graph convolution. The construction of the adjacency matrix $A_i \in \mathbb{R}^{n_i \times n_i}$ involves two key steps: spectral similarity measurement and spatial constraints. First, we calculate the cosine similarity between all pixel pairs

$$S_{jk} = \frac{\mathbf{x}_{ij}^{\top}\mathbf{x}_{ik}}{\|\mathbf{x}_{ij}\| \, \|\mathbf{x}_{ik}\|} \tag{4}$$

where $S_{jk} \in [-1, 1]$ represents the spectral similarity between pixels j and k.

However, a graph constructed based solely on spectral similarity may contain numerous long-range connections. These connections may correspond to pixels that are spatially distant but spectrally similar by chance, which is detrimental to capturing local spatial patterns.

To address this issue, we introduce a spatial window constraint. Specifically, a connection between pixels j and k is considered only when their spatial distance in the original image is less than a preset window radius w. This constraint is implemented via a binary mask matrix $M_i \in \{0, 1\}^{n \times n}$

$$m_{jk} = \begin{cases} 1, & \text{if } \|\mathbf{p}_j - \mathbf{p}_k\|_2 \le w \\ 0, & \text{else} \end{cases} \tag{5}$$

where $\mathbf{p}_j \in \mathbb{R}^2$ denotes the 2-D coordinates of pixel $\mathbf{x}_j$ in the original image, and $\|\cdot\|_2$ represents the Euclidean distance. The final adjacency matrix is obtained by combining the spectral similarity and the spatial constraint

$$A_i = (S_i \odot M_i) + I_n \tag{6}$$

where $\odot$ denotes element-wise multiplication, and I_n is the identity matrix. The addition of the identity matrix I_n is equivalent to introducing self-loops for each node in the graph, i.e., establishing a connection from each pixel to itself. This design ensures that nodes can preserve their own spectral features while aggregating neighborhood information, effectively alleviating the oversmoothing problem that may arise from multilayer graph convolutions.

After constructing the graph structure, LS-GCM employs the message passing neural network (MPNN) framework for feature aggregation. For each node in the graph G_i, we perform the following message-passing operations. First, the original spectral features are used as initial node features. In each message-passing layer, every node aggregates the feature information from its neighboring nodes

$$\mathbf{h}_{ij}^{(l+1)} = \text{ReLU}\left(\mathbf{W}^{(l)}\left[\mathbf{h}_{ij}^{(l)} \,\|\, \sum_{k \in \mathcal{N}(j)} \alpha_{jk}\mathbf{h}_{ik}^{(l)}\right]\right) \tag{7}$$

where $\mathbf{h}_{ij}^{(l)} \in \mathbb{R}^{d_h}$ represents the hidden feature of node j at the lth layer, $\mathcal{N}(j)$ denotes the set of neighboring nodes for node j, $\|$ indicates vector concatenation, ReLU is the ReLU activation function, and $\mathbf{W}^{(l)} \in \mathbb{R}^{d_h \times 2d_h}$ is a learnable weight matrix. The attention weight α_{jk} is obtained through a nonlinear transformation of the edge weights

$$\alpha_{jk} = \frac{\exp\left(\phi\left(a_{jk}\right)\right)}{\sum_{k' \in \mathcal{N}(j)} \exp\left(\phi\left(a_{jk'}\right)\right)} \tag{8}$$

where $\phi : \mathbb{R} \to \mathbb{R}$ is a nonlinear mapping function implemented as a single-layer feedforward network with LeakyReLU activation, i.e., $\phi(x) = \text{LeakyReLU}(w_\phi x + b_\phi)$, which adaptively calibrates the edge weights before softmax normalization, and a_{jk} is the corresponding element value in the adjacency matrix A_i. This attention mechanism allows the model to adaptively adjust the contribution of neighboring nodes based on the importance of the edge. After L layers of message passing, we obtain the final graph-structured embedding $\mathbf{g}_{ij} = \mathbf{h}_{ij}^{(L)} \in \mathbb{R}^{d_g}$ for each node, where d_g is the embedding dimension. Based on the graph-structured embedding $\mathbf{g}_{ij}$, LS-GCM computes the target probability for each pixel through a simple MLP

$$\tilde{z}_{ij} = \sigma\left(\mathbf{W}_z\mathbf{g}_{ij} + b_z\right) \tag{9}$$

where $\mathbf{W}_z \in \mathbb{R}^{k \times d_z}$ and $b_z \in \mathbb{R}$ are learnable parameters, and $\sigma(\cdot)$ is the sigmoid function that maps the output to the interval $[0, 1]$. $\tilde{z}_{ij}$ represents the soft target probability estimate output by the E-step.

LS-GCM plays a critical role within the GSPT-MIL framework. During the E-step, it generates more accurate and consistent target probability estimates by capturing local spatial dependencies between pixels. These estimates serve as pseudo-labels to supervise the detector training in the M-step, thereby breaking the dilemma of label ambiguity in weakly supervised learning. Fig. 2 illustrates the detailed architecture of LS-GCM, which consists of three main components: graph construction, message passing, and target probability estimation. The left part depicts the graph construction process, showing how the adjacency matrix is built from input pixel features through spectral similarity calculation and spatial window constraints. The middle part illustrates the message passing process, featuring multilayer graph convolution operations that include the computation of attention weights. The right part shows the target probability estimation process, demonstrating how target probabilities are generated from graph-structured embeddings via an MLP. Arrows indicate the direction of data flow and the interactions between components. In Section III-C, we will introduce the multiscale contextual position encoder, which works synergistically with LS-GCM to provide global spatial context information for the model.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:51:51 UTC from IEEE Xplore. Restrictions apply.

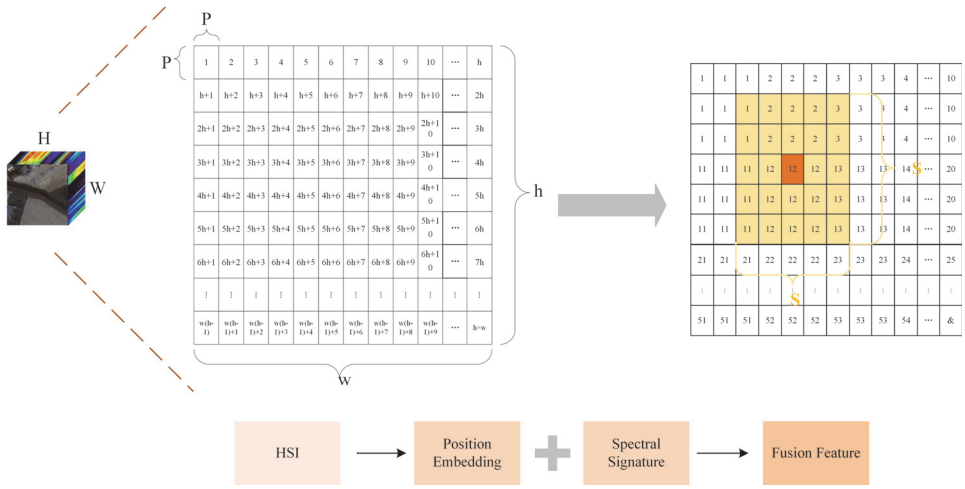


Fig. 3. Architecture diagram of the MCPE.

C. Multiscale Contextual Position Encoder (MCPE)

To address the limitation that spectral features lack absolute spatial awareness, we propose the MCPE module. Models such as Transformers often employ randomly initialized or sinusoidal functions to generate position encodings, which fail to capture the structured spatial semantics of hyperspectral images. MCPE first generates an APS with explicit physical meaning: it partitions the image into a grid and, via a sliding window, counts the contextual frequency of the grid each pixel belongs to, thereby encoding its multiscale spatial position. As illustrated in Fig. 3, this sequence is subsequently projected through a learnable linear layer and fused in an additive manner with the spectral features. This design provides the model with a structured spatial prior ranging from local to global scales, enabling the detector to distinguish between ground objects with similar spectra but different locations, thereby enhancing the representational power of the features.

Unlike conventional position encodings that use fixed sinusoidal functions or learnable yet unstructured embeddings, the core innovation of MCPE lies in proposing a multiscale position encoding strategy capable of simultaneously capturing local details and global layout information. Let the dimensions of the hyperspectral image be $H \times W$, where H is the height and W is the width, resulting in a total of $N_t = H \times W$ pixels. First, we partition the image into regular grid regions. Given a grid size of $P \times P$, the number of grid rows and columns are, respectively,

$$N_h = \lceil H/P \rceil, \quad N_w = \lceil W/P \rceil \tag{10}$$

where $\lceil \cdot \rceil$ denotes the ceiling function, the total number of grids is $N_g = N_h \times N_w$, and each grid is assigned a unique integer label $t \in \{1, 2, \ldots, N_g\}$. For any pixel $\mathbf{x}_{ij}$, corresponding to the spatial location (h_i, w_j) in the image, where $h_i \in \{1, 2, \ldots, H\}$ and $w_j \in \{1, 2, \ldots, W\}$, the index of the grid it belongs to is

$$g_{ij} = \left\lfloor \frac{h_i - 1}{P} \right\rfloor \times N_w + \left\lfloor \frac{w_j - 1}{P} \right\rfloor + 1 \tag{11}$$

where $\lfloor \cdot \rfloor$ denotes the floor function.

Next, we employ a sliding window strategy to construct multiscale contexts. Centered at pixel $\mathbf{x}_{ij}$, a sliding window of size $S \times S$ is defined, where S is an odd number to ensure centrality symmetry. The set of grids covered by this window constitutes the local contextual region for that pixel. Let $\mathcal{W}_{ij}$ denote the set of grids within the sliding window centered at pixel $\mathbf{x}_{ij}$, i.e., $\mathcal{W}_{ij} \subseteq \{1, 2, \ldots, N_g\}$. We count the occurrence frequency of each grid within the window, forming a sparse frequency vector $\mathbf{v}_{ij}$

$$v_{ij}^{(t)} = \text{count}\left(t \in \mathcal{W}_{ij}\right), \quad t = 1, 2, \ldots, N_g \tag{12}$$

where count$(\cdot)$ counts the number of times grid t appears in the window $\mathcal{W}_{ij}$.

Since the sliding window typically covers only a small number of grids, $\mathbf{v}_{ij}$ is a sparse vector with most elements being zero. To reduce dimensionality and enhance representational capacity, we normalize the frequency vector and apply a nonlinear transformation

$$\mathbf{a}_{ij} = \psi\left(\frac{\mathbf{v}_{ij}}{\|\mathbf{v}_{ij}\|_1 + \epsilon}\right) \tag{13}$$

where $\epsilon > 0$ is a small constant to prevent division by zero, $\mathbf{v}_{ij} \in \mathbb{R}^{N_g}$ is the frequency vector of grids within the sliding window, and $\mathbf{a}_{ij} \in \mathbb{R}^{N_g}$ denotes the APS for pixel $\mathbf{x}_{ij}$. $\psi(\cdot)$ is an optional mapping function (set to the identity mapping in this work), reserved as an extension interface for potential dimensionality transformation or nonlinear encoding of the APS in future studies.

After obtaining the APS, MCPE maps it to the same dimensional space as the spectral features through a linear transformation layer, enabling the fusion of positional information with spectral features. Let the spectral features obtained from LS-GCM be denoted as $\mathbf{h}_{ij} \in \mathbb{R}^L$, as described in Section II-B. We project the APS into an L-dimensional space via a linear transformation layer

$$\mathbf{e}_{ij} = \mathbf{W}_p \mathbf{a}_{ij} + \mathbf{b}_p \tag{14}$$

where $\mathbf{W}_p \in \mathbb{R}^{L \times N_e}$ is a learnable weight matrix, $\mathbf{b}_p \in \mathbb{R}^L$ is a bias vector, and $\mathbf{e}_{ij} \in \mathbb{R}^L$ is the resulting position embedding

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:51:51 UTC from IEEE Xplore. Restrictions apply.

vector. The fusion of the position embedding with the spectral features employs a simple additive operation

$$\mathbf{f}_{ij} = \mathbf{h}_{ij} + \mathbf{e}_{ij} \tag{15}$$

where $\mathbf{f}_{ij} \in \mathbb{R}^L$ represents the fused feature vector. This additive fusion strategy is based on the assumption that positional information should be additive with spectral features in the feature space and can be aligned to the same semantic space through linear transformation.

The multiscale characteristics of MCPE are manifested in two aspects. One is the multiscale of the grid size. By adjusting the grid size P, we can control the granularity of the position encoding. A smaller P yields fine-grained position encodings capable of distinguishing subtle positional differences between adjacent pixels; a larger P produces coarse-grained position encodings that focus more on macro-level layout information. In practical applications, multiple grid systems with different P values can be used simultaneously, and the generated multiscale APS can be concatenated or weighted fused to further enhance the representational power of the position encoding. The other aspect is the multiscale of the sliding window size. The sliding window size S controls the scope of the local context. A smaller S makes the position encoding focus more on the immediate context, suitable for capturing local patterns; a larger S encompasses broader contextual information, aiding in understanding the global spatial structure. Similarly, multiple sliding windows with different S values can be employed to generate multiscale contextual features.

MCPE plays a significant role within the GSPT-MIL framework. During the M-step, it injects absolute positional information into the feature learning process, enabling the detector to make decisions by leveraging both spectral characteristics and spatial location. This position-aware capability is crucial for distinguishing materials with similar spectra but different spatial positions, especially in complex scenes. Fig. 3 illustrates the detailed architecture of MCPE, which consists of three main components: grid partitioning, sliding window statistics, and positional encoding fusion. The upper left part shows a schematic of grid partitioning, demonstrating how the image is divided into regular grids and numbered; the upper right part shows a schematic of sliding window statistics, illustrating how a sliding window centered on the target pixel covers multiple grids; the lower part depicts the positional encoding fusion process, showing the workflow from the original image to feature fusion, with arrows indicating the direction of data flow and interactions between components. In Section II-D, we will introduce the SS-FAN, which further optimizes the synergistic representation of spectral features and positional features.

D. Spectral–Spatial Feature Alignment Network

To achieve the deep fusion of two heterogeneous modalities—spectral and positional information—we introduce the SS-FAN. Given that simple feature concatenation or addition can hardly guarantee the semantic consistency between the two feature types, SS-FAN employs a contrastive learning mechanism. It treats the spectral features and positional features of the same pixel as a positive sample pair, maps them into a shared subspace via projection networks, and subsequently maximizes their mutual information. As shown in Fig. 4, this module functions as a lightweight, plug-and-play unit, optimized jointly with the classification loss during the M-step. By aligning the two feature representations, it yields features that are more discriminative and robust to noisy pseudo-labels, thereby enhancing the model's detection accuracy and robustness in complex scenes.

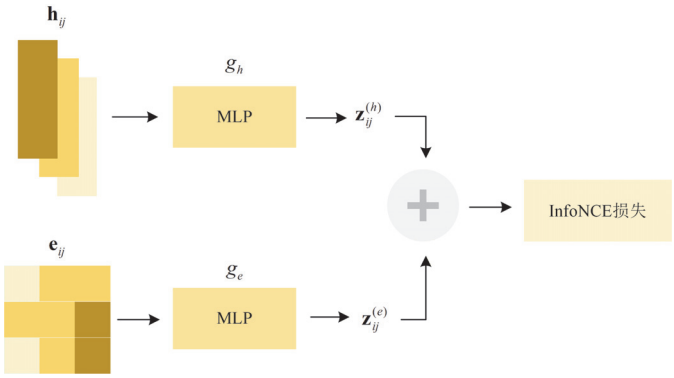


Fig. 4. Architecture diagram of the SS-FAN.

SS-FAN is built upon the feature extraction process of the M-step. As described in Section II-C, for a given pixel $\mathbf{x}_{ij}$, we obtain a fused feature $\mathbf{f}_{ij}$ by combining its spectral feature $\mathbf{h}_{ij}$ and position embedding feature $\mathbf{e}_{ij}$. The initial additive fusion assumes that spectral and positional information are linearly combinable in the feature space. However, this simple fusion approach may not fully capture the deep correlations between the two modalities. Therefore, SS-FAN introduces a more sophisticated feature interaction and alignment mechanism. The core idea of SS-FAN is to leverage contrastive learning to pull closer the distances between the spectral feature and position feature of the same pixel in a projected space, while pushing apart the distances between features from different pixels. First, we define two feature projection functions to map the spectral feature $\mathbf{h}_{ij}$ and the position embedding $\mathbf{e}_{ij}$ into a shared contrastive learning space $\mathbb{R}^{d_e}$

$$\mathbf{z}_{ij}^{(h)} = g_h\left(\mathbf{h}_{ij}\right) = \mathbf{W}_h^{(2)}\mathrm{ReLU}\left(\mathbf{W}_h^{(1)}\mathbf{h}_{ij} + \mathbf{b}_h^{(1)}\right) + \mathbf{b}_h^{(2)} \tag{16}$$

$$\mathbf{z}_{ij}^{(\epsilon)} = g_\epsilon\left(\mathbf{e}_{ij}\right) = \mathbf{W}_e^{(2)}\mathrm{ReLU}\left(\mathbf{W}_e^{(1)}\mathbf{e}_{ij} + \mathbf{b}_e^{(1)}\right) + \mathbf{b}_e^{(2)}. \tag{17}$$

Here, $g_h(\cdot)$ and $g_\epsilon(\cdot)$ are parameter-independent MLP projection heads; $\mathbf{W}_h^{(1)}, \mathbf{W}_e^{(1)} \in \mathbb{R}^{d_m \times L}$ and $\mathbf{W}_h^{(2)}, \mathbf{W}_e^{(2)} \in \mathbb{R}^{d_c \times d_m}$ are weight matrices; $\mathbf{b}_h^{(1)}, \mathbf{b}_e^{(1)} \in \mathbb{R}^{d_m}$ and $\mathbf{b}_h^{(2)}, \mathbf{b}_e^{(2)} \in \mathbb{R}^{d_c}$ are bias vectors; ReLU is the ReLU activation function; d_m is the hidden layer dimension.

Next, we construct positive and negative sample pairs for contrastive learning. For pixel $\mathbf{x}_{ij}$, its projected pair $(\mathbf{z}_{ij}^{(h)}, \mathbf{z}_{ij}^{(e)})$ constitutes a positive sample pair, as they originate from different modalities (spectral and positional) of the same pixel. The projection vectors of all other pixels within a batch $\mathcal{B}$ serve as negative samples. We adopt the InfoNCE loss as

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:51:51 UTC from IEEE Xplore. Restrictions apply.

the objective function for contrastive learning. For a positive sample pair, the loss is defined as

$$\mathcal{L}_{ij}^{(h,e)} = -\log\left(\exp\left(s\left(\mathbf{z}_{ij}^{(h)}, \mathbf{z}_{ij}^{(e)}\right)/\tau\right)/ \times\left[\sum_{k=1}^{B}\exp\left(s\left(\mathbf{z}_{ij}^{(h)}, \mathbf{z}_{ik}^{(e)}\right)\Big/\tau\right) + \sum_{k=1}^{B}\exp\left(s\left(\mathbf{z}_{ij}^{(h)}, \mathbf{z}_{ik}^{(h)}\right)\Big/\tau\right)\right] \tag{18}$$

where $s(\mathbf{u}, \mathbf{v}) = \mathbf{u}^{\top}\mathbf{v}/(\|\mathbf{u}\|\|\mathbf{v}\|)$ denotes cosine similarity. The summation term in the denominator traverses all pixels within the batch (including itself), encompassing all possible negative sample pairs. For symmetry, we also compute the loss with the positional feature as the anchor

$$\mathcal{L}_{ij}^{(e,h)} = -\log\left(\exp\left(s\left(\mathbf{z}_{ij}^{(e)}, \mathbf{z}_{ij}^{(h)}\right)/\tau\right)/ \times\left[\sum_{k=1}^{B}\exp\left(s\left(\mathbf{z}_{ij}^{(e)}, \mathbf{z}_{ik}^{(h)}\right)\Big/\tau\right) + \sum_{k=1}^{B}\exp\left(s\left(\mathbf{z}_{ij}^{(e)}, \mathbf{z}_{ik}^{(e)}\right)\Big/\tau\right)\right]. \tag{19}$$

The overall contrastive loss of SS-FAN is the average of the losses over all positive sample pairs within the batch

$$\mathcal{L}_{align} = \frac{1}{B}\sum_{i=1}^{B}\frac{1}{2}\left(\mathcal{L}_{ij}^{(h,e)} + \mathcal{L}_{ij}^{(e,h)}\right). \tag{20}$$

The SS-FAN module is not trained independently; instead, it is jointly optimized with the classification objective of the M-step. The total loss function for the M-step is defined as

$$\mathcal{L}_M = \mathcal{L}_{cls} + \alpha \cdot \mathcal{L}_{align} \tag{21}$$

where

$$\mathcal{L}_{cls} = -\frac{1}{B}\sum_{i=1}^{B}\left[\hat{z}_{ij}\log p_{ij} + \left(1-\hat{z}_{ij}\right)\log\left(1-p_{ij}\right)\right]$$

is the binary cross-entropy classification loss, $\hat{z}_{ij}$ is the pseudo-label generated by the E-step, p_{ij} is the classifier's predicted probability for the fused feature $\mathbf{f}_{ij}$, and $\alpha > 0$ is the weight coefficient balancing the two losses.

This joint optimization mechanism ensures that the feature alignment process consistently serves the ultimate classification objective. The contrastive loss $\mathcal{L}_{\text{align}}$ acts as a regularization term, encouraging the model to learn the consistency between the spectral and positional modalities, thereby producing feature representations that are more discriminative and robust to noisy pseudo-labels. Fig. 4 illustrates the workflow and architectural details of SS-FAN. First, the feature inputs are shown, displaying the spectral feature $\mathbf{h}_{ij}$ and positional feature $\mathbf{e}_{ij}$ from the M-step. The middle part represents the projection heads, where two independent MLPs (g_h and g_e) project the input features into a shared contrastive space, yielding $\mathbf{z}_{ij}^{(h)}$ and $\mathbf{z}_{ij}^{(e)}$, respectively. The right side depicts the contrastive loss computation, illustrating the symmetric InfoNCE loss calculated based on positive and negative sample pairs, which guides the model update in the M-step. So far, we have completed the elaboration of the three core modules of GSPT-MIL. Next, Section II-E will integrate these modules and provide a detailed description of the complete EM optimization algorithm workflow and its implementation details.

E. Weakly Supervised Training Based on EM

The core training framework of the GSPT-MIL model is built upon the classical EM algorithm. This framework enables effective learning of a pixel-level object detector under only bag-level labels by alternating between the E-step (target position inference) and the M-step (detector parameter learning). This section will elaborate on the specific implementation and optimization objectives of this EM training process.

In the E-step, the objective of the model is to infer the probability that each pixel belongs to the target based on the current state of the object detector, i.e., to estimate the posterior distribution of the latent variable z_{ij}. This step is primarily conducted by the LS-GCM, whose input is the spectral features of all pixels within the data bag B_i, and whose output is the soft target label $\tilde{z}_{ij}$ for each pixel.

The specific procedure consists of four steps. The first step is feature preparation. For a data bag B_i, the deep spectral feature $\mathbf{h}_{ij}$ of each pixel is first extracted using the spectral feature extractor obtained from the current M-step training. These features serve as the input node features for the LS-GCM. The second step is graph structure embedding generation. The LS-GCM module constructs a local similarity graph according to (4)–(6) and aggregates neighborhood information through the message passing process defined by (7) and (8) to generate the graph structure embedding $\mathbf{g}_{ij}$ for each pixel. The third step is soft label calculation. As shown in Fig. 2, the graph structure embedding $\mathbf{g}_{ij}$ is transformed into a target probability estimate $\tilde{z}_{ij}$ through an MLP and a sigmoid activation function, as shown in (9). The obtained $\tilde{z}_{ij}$ is a continuous soft label, reflecting the confidence that the pixel is considered a target by the current model. The fourth step is pseudo-label generation. To provide clear supervision signals for the M-step, the soft labels need to be binarized into pseudo-labels $\hat{z}_{ij}$. We innovatively introduce a dynamic threshold strategy to address the variations in target distribution across different positive bags. For each positive bag, its dynamic threshold τ_i is calculated as

$$\tau_i = \min_j \tilde{z}_{ij} + \lambda\left(\max_j \tilde{z}_{ij} - \min_j \tilde{z}_{ij}\right) \tag{22}$$

where $\lambda \in [0, 1]$ is a hyperparameter that controls the strictness of the threshold.

Subsequently, pseudo-labels for supervising the M-step are generated

$$\hat{z}_{ij} = \begin{cases} 1, & \text{if } \tilde{z}_{ij} \geq \tau_i \text{ and } L_i = 1 \\ 0, & \text{otherwise.} \end{cases} \tag{23}$$

For negative bags, all $\hat{z}_{ij}$ are directly set to 0.

The training objective of the E-step is to align the soft label $\tilde{z}_{ij}$ generated by LS-GCM with the probability p_{ij} predicted by the current M-step detector as closely as possible.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:51:51 UTC from IEEE Xplore. Restrictions apply.

Accordingly, the binary cross-entropy loss for the E-step is defined as

$$\mathcal{L}_E = -\frac{1}{N_{\text{bag}}} \sum_{i=1}^{K} \sum_{j=1}^{n_i} \left[p_{ij} \log\left(\tilde{z}_{ij}\right) + \left(1 - p_{ij}\right) \log\left(1 - \tilde{z}_{ij}\right) \right] \tag{24}$$

where N_{bag} denotes the total number of pixels within a batch. Minimizing $\mathcal{L}_E$ enables LS-GCM to learn to incorporate spatial context information and produce smoother and more consistent target position estimates compatible with the knowledge of the current detector.

In the M-step, the model aims to optimize the parameters of the object detector by utilizing the pseudo-labels generated in the E-step as supervision signals. This step integrates the multiscale contextual position encoder and the SS-FAN to learn powerful feature representations that fuse spectral and spatial information.

The specific process consists of three steps. The first step is spectral–spatial feature extraction and fusion. For a given pixel $\mathbf{x}_{ij}$, in step M, spectral features $\mathbf{h}_{ij}$ are first extracted using a 1D-CNN, while its APS $\mathbf{a}_{ij}$ is generated through MCPE and subsequently projected to obtain position embeddings $\mathbf{e}_{ij}$. Following this, feature fusion is performed according to (15) to derive $\mathbf{f}_{ij}$. The second step is target classification, where the fused feature $\mathbf{f}_{ij}$ is fed into a classifier to predict the posterior probability p_{ij} of the pixel belonging to the target. The final step is feature alignment. As illustrated in Fig. 4, the SS-FAN module receives $\mathbf{h}_{ij}$ and $\mathbf{e}_{ij}$ simultaneously. It maps these inputs into a contrastive space via projection heads defined by (16) and (17) and employs the symmetric contrastive loss $\mathcal{L}_{\text{align}}$, as specified in (18)–(20), to promote the mutual alignment of the spectral and positional features of the same pixel within the latent space.

The complete training process of GSPT-MIL is an optimization procedure involving alternating iterations between the E-step and the M-step. Its global optimization objective can be regarded as the indirect maximization of the log-likelihood of the observed data.

1) Parameter Initialization and Warm-up Stage: After random initialization of the model parameters, a brief warm-up stage is first conducted. In this stage, the pseudo-labels for the M-step are derived directly from the bag-level labels without performing the E-step. This facilitates the detector in acquiring preliminary discriminative capability, providing a favorable starting point for subsequent EM iterations.

2) EM Alternating Iteration: Upon completion of the warm-up stage, the formal EM alternating iteration begins. First, the detector is held fixed, and the LS-GCM is updated (E-step): With the parameters of the M-step detector fixed, the soft targets p_{ij} predicted by the current detector are utilized to update the parameters ϕ of the LS-GCM by minimizing the loss $\mathcal{L}_E$, thereby generating new pseudo-labels $\hat{z}_{ij}$. Next, with the parameters ϕ of the LS-GCM held fixed, the detector and the SS-FAN are updated (M-step). Leveraging the newly generated pseudo-labels $\hat{z}_{ij}$, the parameters θ of the M-step detector and the SS-FAN are updated by minimizing the loss $\mathcal{L}_M$.

3) Model Convergence: Step 2) is repeated until the model's performance on the validation set no longer improves or the preset maximum number of iterations is reached. After each iteration round, the position inference in the E-step and the detection capability in the M-step mutually enhance each other, allowing the quality of pseudo-labels and model performance to co-evolve.

Through this meticulously designed EM training framework, GSPT-MIL successfully achieves effective training of a pixel-level object detector solely on weak bag-level labels. Experiments demonstrate that this framework converges stably and significantly outperforms existing weakly supervised methods.

III. Experiments

A. Experimental Setup

To comprehensively and fairly evaluate the detection performance of the proposed GSPT-MIL method, we selected nine representative state-of-the-art algorithms as comparison benchmarks. These methods encompass different technical routes in the field of weakly supervised multiple-instance hyperspectral target detection, ranging from traditional concept learning to deep neural networks. Specifically, the MIL methods based on target concept learning include EM-DD [33], MILES [25], MI-HE [21], MI-ACE [17], MI-SMF [17], MIMD-ACE [22], and MIMD-SMF [22]. These methods construct detectors by learning target and background prototypes at the bag level and serve as classic and widely adopted baseline approaches. To evaluate the performance improvements brought by deep feature extraction and end-to-end optimization, we also selected two advanced MIL methods based on deep neural networks, namely Attention-DMIL [29] and VMIL-ECM [32]. The latter, sharing a similar EM framework for weakly supervised detection, serves as a more direct comparison baseline with our method. All comparison methods were implemented using publicly available source code from the authors or carefully reproduced following the original papers, with their optimal parameter settings adopted to ensure fairness and reliability of the comparison.

Our GSPT-MIL model is implemented using the PyTorch framework. The model parameters are set as described in Section II: grid size $P = 5$, sliding window size $S = 7$, graph neighborhood radius $w = 5$, feature dimension $d_h = 256$, dynamic threshold coefficient $\lambda = 0.25$, and contrastive loss weight $\alpha = 0.1$. The Adam optimizer is employed for training, with an initial learning rate of 5×10^{-4} and a cosine annealing strategy for learning rate decay. The batch size is set to $B = 128$, which denotes the number of bags randomly sampled in each iteration. For positive bags, we randomly sample a fixed number of pixels to ensure balanced representation, while for negative bags, all pixels are treated as background instances. The total number of training epochs is 2000, with a warm-up epoch number of $W = 200$. An early stopping strategy is adopted during training, terminating the process when the validation set performance does not improve for 50 consecutive epochs. To ensure the statistical reliability of the results, each experiment is independently run five times on randomly partitioned datasets, and the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:51:51 UTC from IEEE Xplore. Restrictions apply.

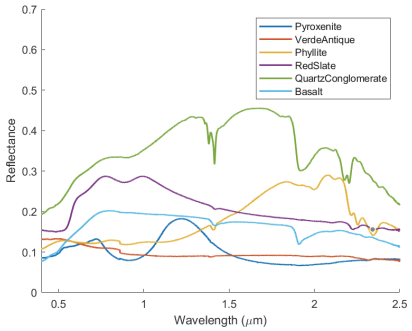


Fig. 5. Six endmembers selected from the ASTER spectral library. Pyroxenite is designated as the target of interest, and the remaining five spectra serve as background endmembers.

average performance along with the standard deviation is reported.

We adopt the area under the receiver operating characteristic curve area under the ROC curve (AUC) as the primary performance evaluation metric. The ROC curve characterizes the relationship between the true positive rate (TPR) and the false positive rate (FPR) of a detector under varying thresholds. To comprehensively evaluate performance, we report two types of detection metrics depending on the dataset. For simulated datasets, we use the conventional AUC, which computes the AUC over the entire FPR interval from 0 to 1, providing a holistic assessment of the detector's overall performance. For real-world datasets, we employ the normalized AUC (NAUC), which calculates the NAUC within the low false alarm rate interval of $[0, 0.001]$. This metric better reflects detection performance under low false alarm conditions, which are of greater concern in practical applications. In addition, we present target-background separation plots for visual analysis, intuitively illustrating the discriminative capability of different methods in terms of the response values between targets and background.

B. Simulated Dataset Experiments

To validate the fundamental detection capability of GSPT-MIL under controlled conditions and its robustness to varying target abundances, we first conducted experiments on three simulated datasets. These datasets were generated using six spectra selected from the ASTER spectral library, as illustrated in Fig. 5. The target, pyroxenite, was embedded as subpixel components within a complex background, while the remaining five candidate spectra served as background endmembers for generating mixed background pixels and subpixel targets. Finally, samples were generated using a linear mixing model. The signal-to-noise ratio of the simulated datasets was set to 30 dB. Each dataset comprised 1000 bags, including 300 positive bags and 700 negative bags, with each positive bag containing five subpixel targets. We set the average target abundance ratios to 0.1, 0.2, and 0.3, respectively, simulating a variety of real-world scenarios with target abundances ranging from extremely low to moderate.

Table I presents the AUC performance of all compared methods on the three simulated datasets. Overall, as the average target abundance ratio $\bar{p}_t$ increases from 0.1 to 0.3, the detection performance of all methods exhibits an upward trend, attributable to the enhanced target signal that facilitates easier separation from the background. However, across different abundance ratios, GSPT-MIL consistently achieves the best performance, with AUC values significantly and stably outperforming all compared methods.

TABLE I

AUC OF DIFFERENT ALGORITHMS UNDER VARYING TARGET ABUNDANCE RATIOS ON THE SIMULATED DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Algorithm	$\bar{p}_t$			Avg. AUC
	0.1	0.2	0.3	
EM-DD	0.645	0.727	0.825	0.732
MILES	0.737	0.772	0.775	0.761
MI-HE	0.699	0.779	0.816	0.765
MI-ACE	0.682	0.744	0.843	0.756
MI-SMF	0.684	0.745	0.845	0.758
MIMD-ACE	0.654	0.754	0.830	0.746
MIMD-SMF	0.661	0.739	0.838	0.746
Attention-DMIL	0.729	0.823	0.846	0.799
VMIL-ECM	0.794	0.866	0.897	0.852
GSPT-MIL	**0.810**	**0.874**	**0.918**	**0.867**

As shown in the table, the performance of concept learning-based MIL methods is relatively low. Particularly under the lowest target abundance ratio of 0.1, the AUC of most methods falls below 0.7. This indicates that, in scenarios involving subpixel targets and complex mixed backgrounds, relying on a single target spectrum or learning limited prototypes is insufficient to adequately model the subtle variations of the target and the diversity of the complex background. Deep learning-based methods generally outperform concept learning-based approaches, validating the advantages of deep feature extraction. Notably, VMIL-ECM, the baseline most closely related to our method, achieves strong performance through its EM framework and variational inference, attaining an AUC of 0.897 at an abundance ratio of 0.3. This demonstrates that deep networks can learn more discriminative feature representations and better capture the subtle differences between targets and the background. The proposed GSPT-MIL method achieves an average AUC approximately 1.5 percentage points higher than the second-best method across the three datasets, with this advantage being most pronounced at the abundance ratio of 0.3. This is attributed to the fact that our method explicitly models spatial dependencies among pixels through LS-GCM, leveraging the local clustering cues of subpixel targets. Furthermore, the incorporation of absolute positional information via MCPE enhances the spatial discriminability of features. Finally, the combination of contrastive alignment via SS-FAN and dynamic pseudo-label optimization within the EM framework collectively enables more accurate and robust detection of subpixel targets under low signal-to-noise conditions.

To more intuitively illustrate the performance differences, we take the dataset with an average target abundance ratio of 0.3 as an example and plot the ROC curves of the main comparison methods (Fig. 6) along with the boxplots of target-background response separability (Fig. 7). As can be seen from the ROC curves, GSPT-MIL exhibits the curve closest to the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:51:51 UTC from IEEE Xplore. Restrictions apply.

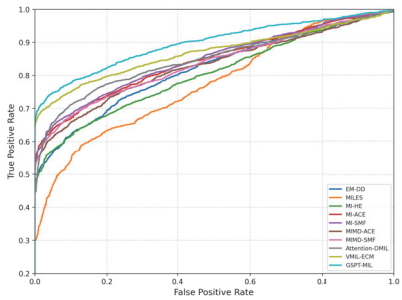


Fig. 6. ROC curve comparison on the simulated dataset ($\bar{P}_t = 0.3$).

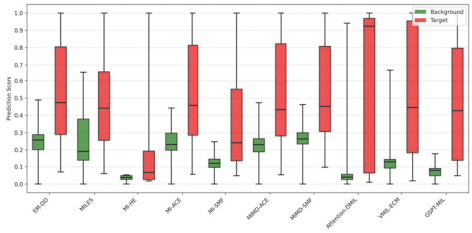


Fig. 7. Statistical separability analysis on the simulated dataset ($\bar{P}_t = 0.3$).

top-left corner, maintaining the highest TPR across the entire FPR range, with a particularly prominent advantage in the low FPR region—a critical factor for practical applications that demand low false alarm rates. Furthermore, the statistical separability plots further reveal the intrinsic capability of different methods in distinguishing between targets and background. Fig. 7 presents the distributions of detection statistics output by each method for all target pixels and background pixels. In the boxplots, the upper, middle, and lower lines represent the maximum, median, and minimum values, respectively, while the box range denotes the interquartile range. For an ideal detector, the target and background boxes should be completely separated, with as large a gap as possible. From Fig. 7, we observe that methods such as MI-HE and Attention-DMIL achieve relatively good background suppression; however, their target responses are comparatively weak, and the separation between target and background distributions is limited, with noticeable overlap. In contrast, the proposed GSPT-MIL maintains strong background suppression capability, and the gap between the red and green boxes is nearly larger than that of any comparable method, thereby achieving a substantial target-background separation margin. This outcome can be directly attributed to our method's ability to obtain high-quality pseudo-labels through accurate location inference and to leverage features enriched with extensive spatial context information to train a highly discriminative detector. These results intuitively validate that GSPT-MIL, through precise location inference and discriminative feature learning, effectively enhances target signals while suppressing background noise.

C. Experiments on the MUUFL Gulfport Hyperspectral Dataset

After validating the fundamental detection capability of GSPT-MIL on simulated datasets, we further conducted

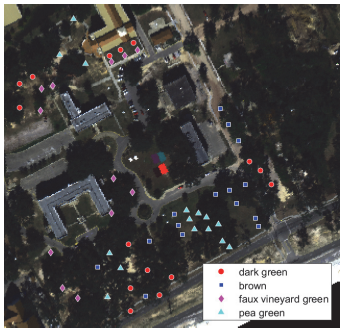


Fig. 8. MUUFL Gulfport dataset and ground-truth target locations.

TABLE II
TARGET INFORMATION FOR THE MUUFL GULFPORT DATASET

Target Type	Num. of Targets with Different Sizes		
	$0.5 \times 0.5\,\mathrm{m}^2$	$1 \times 1\,\mathrm{m}^2$	$3 \times 3\,\mathrm{m}^2$
Brown	5(m)	5(m)	5(m)
Dark Green	5(m)	5(m)	5(m)
Faux Vineyard Green	4(m)	4(m)	4(m)
Pea Green	5(m)	5(m)	5(m)
All Types	19(m)	19(m)	19(m)

experiments on the more challenging real-world MUUFL Gulfport dataset [34] to evaluate the model's performance under complex practical conditions. This dataset was acquired in November 2010 over the University of Southern Mississippi campus. Five hyperspectral images were collected at this site, each comprising approximately 330 × 340 pixels with a spatial resolution of 1 m. Within the wavelength range of 375–1050 nm, there are 72 bands with a spectral sampling interval of approximately 10 nm; we removed eight bands, specifically bands 1–4 and 69–72. The scene contains four cloth targets of different colors, some of which are partially obscured by oak trees, representing a typical subpixel detection task. Fig. 8 shows a color image of the scene along with the ground-truth target locations. In the experiment, we selected Flight 1 and Flight 3 images for model training and used Flight 4 for testing. Data bags were constructed based on GPS information, with the GPS equipment having an accuracy of approximately 1–2 m. For each target, we delineated a 5 × 5 region around the corresponding GPS coordinates as a positive bag, ensuring the target was contained within this region. Regions without any target types were designated as negative bags.

We first evaluate the detection performance of the model on four different types of targets—brown (BR), dark green (DG), faux vineyard green (FVG), and pea green (PG)—with the four target types listed in Table II. The Bullwinkle algorithm [35] is employed to generate the ROC curves, and the NAUC at an FPR of 0.001 [17], [21], [22], [23] is adopted as the primary evaluation metric, which better reflects detection performance under low false alarm conditions in practical applications. Table III presents the NAUC values of all compared methods for the four target types, while Fig. 9 shows the corresponding ROC curves at an FPR of 0.001.

Overall, the performance of concept learning-based MIL methods is generally low, with most average NAUC values

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:51:51 UTC from IEEE Xplore. Restrictions apply.

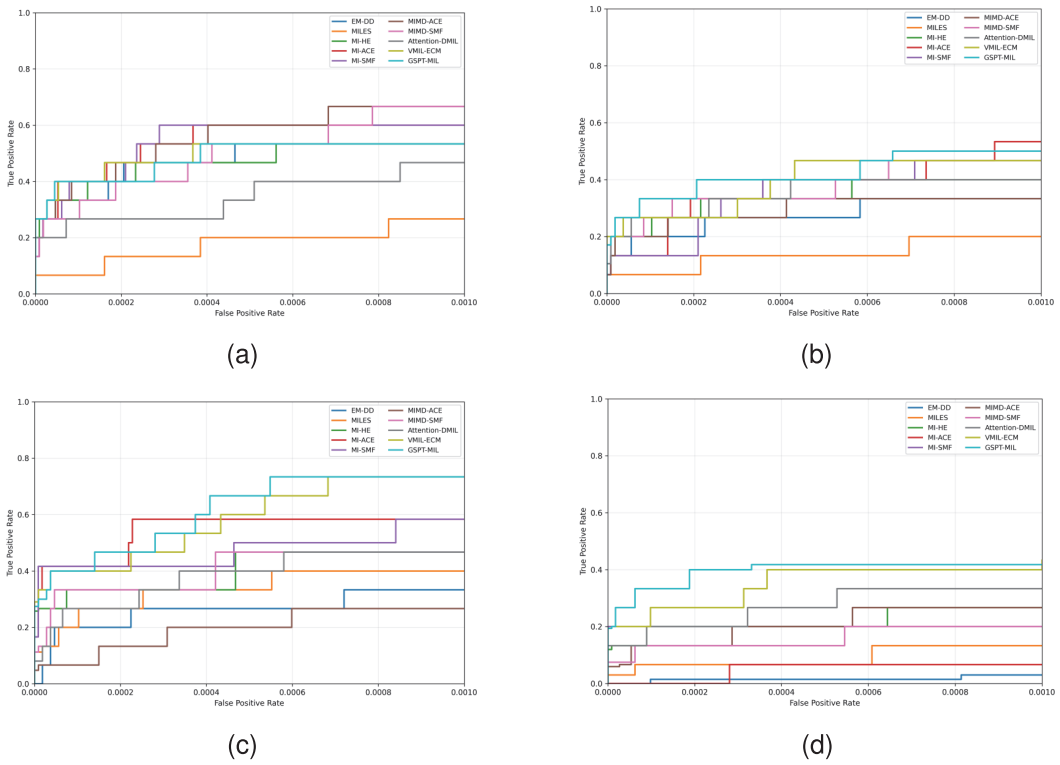


Fig. 9. ROC curves on the MUUFL Gulfport dataset at an FPR of 0.001. (a) Brown. (b) DG. (c) FVG. (d) PG.

TABLE III
TARGET DETECTION PERFORMANCE OF DIFFERENT ALGORITHMS ON THE MUUFL GULFPORT DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Alg.	Brown	Dark Green	Faux Vineyard Green	Pea Green	Avg.NAUC
EM-DD	0.471	0.290	0.283	0.037	0.270
MILES	0.207	0.147	0.376	0.069	0.120
MI-HE	0.529	0.356	0.456	0.226	0.392
MI-ACE	0.539	0.369	0.543	0.048	0.375
MI-SMF	**0.540**	0.349	0.472	0.200	0.390
MMD-ACE	0.539	0.322	0.218	0.223	0.326
MMD-SMF	0.516	0.385	0.411	0.180	0.373
Attention-DMIL	0.361	0.341	0.382	0.273	0.339
VMIL-ECM	0.488	0.407	0.581	0.372	0.462
GSPT-MIL	0.501	**0.419**	**0.596**	**0.395**	**0.478**

falling below 0.45. This indicates that in real-world complex scenarios, strategies relying on a single target spectrum or learning limited prototypes struggle to cope with challenges such as target diversity, spectral variability, and complex background interference. The subpixel mixing caused by partial target occlusion by trees further exacerbates this difficulty. Deep learning-based methods such as Attention-DMIL and VMIL-ECM significantly outperform traditional approaches, demonstrating the powerful capabilities of deep learning in feature extraction and complex pattern learning. Notably, VMIL-ECM, serving as the baseline model for our method, achieves strong performance through its EM framework and variational inference, attaining an average NAUC of 0.462. The proposed GSPT-MIL achieves the best performance on the DG, FVG, and PG target types, with an average NAUC of 0.478 across the four target types, outperforming VMIL-ECM by approximately 1.6 percentage points. This advantage is particularly pronounced on the most challenging DG and FVG targets.

To intuitively demonstrate the discriminative capability of different methods in distinguishing targets from background, Fig. 10 presents the statistical separability boxplots for the four single-target detection tasks. The following observations can be drawn from the figure. Methods such as MI-HE and Attention-DMIL exhibit good background suppression performance on some targets. However, this strong background suppression often comes at the cost of reduced target responses, resulting in insufficient target-background separation. Compared with traditional methods, VMIL-ECM significantly suppresses background responses, but target responses also decrease to some extent, leading to limited improvement in separability. GSPT-MIL achieves strong target-background separation across all four target types, characterized by relatively high median target responses, low

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:51:51 UTC from IEEE Xplore. Restrictions apply.

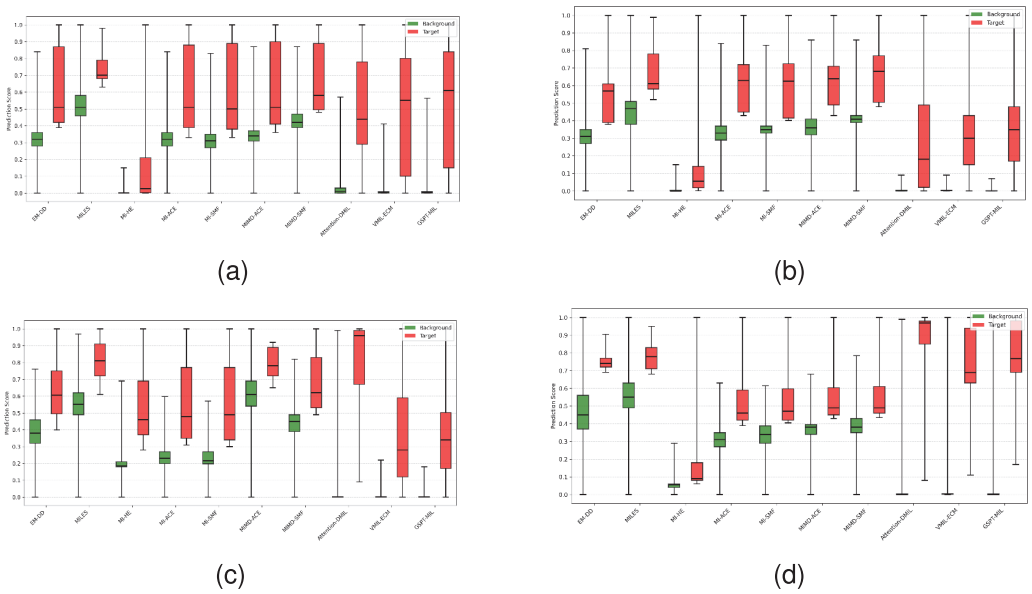


Fig. 10. Statistical separability analysis on the MUUFL Gulfport dataset. (a) Brown. (b) DG. (c) FVG. (d) PG.

median background responses, and the most compact background distribution. These results intuitively validate that GSPT-MIL, through precise target localization in the E-step and discriminative feature learning in the M-step, effectively enhances target signals while suppressing background noise.

D. Ablation Study

To validate the contributions of each core module in GSPT-MIL, we designed multiple ablation variants and compared their performance on the simulated dataset with a target abundance ratio of 0.3 and on the Brown target detection task using the MUUFL Gulfport dataset. All variants maintained the same settings as described above. We constructed seven network variants and evaluated them using a controlled variable approach, ensuring that all configurations except for the ablated components remained identical to guarantee fair comparison.

Among the variants, Variant 1 removes LS-GCM; the E-step no longer aggregates neighborhood information via the graph structure, but directly feeds the spectral features extracted by the 1D-CNN into an MLP to generate soft labels, thereby verifying the contribution of graph-structured spatial dependency modeling to target localization accuracy. Variant 2 removes MCPE; the M-step does not generate absolute positional sequences or their embeddings, and relies solely on spectral features for classification, aiming to assess whether absolute positional information aids in distinguishing pixels with similar spectra but different spatial locations. Variant 3 removes SS-FAN; the M-step eliminates the contrastive learning loss and optimizes the detector solely based on the pseudo-label classification loss, to examine the effectiveness of cross-modal feature alignment. Variant 4 removes the EM alternating training; the model undergoes only the warm-up phase, after which the initial state of the E-step is fixed, and the M-step is trained exclusively with the fixed pseudo-labels generated during warm-up, to validate the critical role of the collaborative iterative optimization between the E-step and M-step in sustaining performance improvement. Variant 5 removes both LS-GCM and MCPE, using neither graph structure modeling nor absolute positional encoding, and performs classification solely based on the spectral features extracted by the 1D-CNN, serving as the simplest baseline to highlight the joint importance of spatial information. Variant 6 replaces LS-GCM with a 1-D convolution along the pixel dimension; this fixed convolutional kernel method has fewer parameters and is used to evaluate whether the dynamic graph structure captures nonlinear dependencies and long-range relationships among pixels more effectively than fixed-kernel convolutions. Variant 7 substitutes MCPE with the standard sinusoidal positional encoding, which relies solely on sequence order rather than spatial grid statistics, to verify whether the multi-scale contextual counting approach based on spatial statistics outperforms conventional frequency-based encoding methods.

From Table IV, it can be observed that removing LS-GCM leads to a significant performance drop, with a decrease of 1.2 in AUC on the simulated dataset and 3.0% in NAUC on the Gulfport dataset. This indicates that graph-based modeling plays an irreplaceable role in capturing local spatial correlations and improving target localization accuracy. Particularly in real-world scenarios, where targets often appear as clusters of pixels, LS-GCM aggregates neighborhood information through message passing, effectively enhancing the signal strength of weak targets while suppressing interference from isolated noise. These results verify our motivation for introducing graph convolutions, namely to leverage spatial dependencies under the non-i.i.d. assumption. Notably, the performance degradation of LS-GCM is more pronounced on the real-world dataset than on the simulated dataset, reflecting that complex real-world environments rely more heavily on spatial context modeling. When background noise increases and target boundaries become ambiguous, the capability of graph structures to aggregate neighborhood information becomes critical

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:51:51 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
QUANTITATIVE ANALYSIS OF DIFFERENT VARIANTS IN THE ABLATION STUDY

Methods	Simulated Dataset ($\bar{p}_t = 0.3$) AUC	MUUFL Gulfport (brown) NAUC
w/o LS-GCM (Variant 1)	0.907	0.486
w/o MCPE (Variant 2)	0.911	0.493
w/o SS-FAN (Variant 3)	0.915	0.497
w/o EM Alternation (Variant 4)	0.891	0.355
w/o Global Context (Variant 5)	0.899	0.474
LS-GCM replaced by 1D-CNN (Variant 6)	0.910	0.491
MCPE replaced by Sinusoidal PE (Variant 7)	0.916	0.497
Full Model	0.918	0.501

for stable detection. Variant 2 exhibits decreases of 0.8% and 1.6% on the simulated and real-world datasets, respectively, demonstrating that absolute position encoding provides essential spatial context for the model. Especially in complex backgrounds where spectral features may be highly ambiguous, the absolute positional information of pixels becomes a key clue for distinguishing targets from background. The position sequences generated by MCPE through multiscale sliding windows assign unique spatial identifiers to each pixel, significantly enhancing feature discriminability. The more substantial decline on the real-world dataset suggests that when spectral separability is reduced, the compensatory role of positional prior information in model inference becomes increasingly important. Removing SS-FAN results in a decrease of 0.3% in AUC on the simulated dataset and 0.8% in NAUC on the real-world dataset, indicating that the contrastive learning mechanism effectively promotes the alignment of spectral features and spatial position features in the latent space. This alignment not only enhances feature robustness but also provides additional regularization through self-supervised signals, helping to prevent overfitting under weakly supervised conditions. The experimental results confirm the effectiveness of cross-modal contrastive learning, demonstrating that the design of SS-FAN is not redundant. Although the magnitude of the performance drop is relatively modest, considering that SS-FAN primarily facilitates feature alignment rather than direct spatial modeling, its contribution to final performance remains positive and necessary.

Removing the alternating EM training mechanism, where the model relies solely on the fixed pseudo-labels obtained during the warm-up phase for M-step training, results in a 2.9% decrease in AUC on the simulated dataset and a dramatic 29.1% drop in NAUC on the real-world dataset—the most severe performance degradation among all variants. This outcome demonstrates that the collaborative iteration between the E-step and M-step is critical for continuous model optimization. There are three primary reasons why the warm-up phase alone is insufficient. First, the pseudo-labels generated during warm-up are directly propagated from bag-level labels rather than inferred by the detector itself. This coarse assignment deviates significantly from the true target distribution, forcing the detector to be trained under noisy supervisory signals. Second, owing to the absence of subsequent E-step updates, these erroneous pseudo-labels remain fixed throughout training, causing the detector to converge to a suboptimal solution biased by the initial noise. Its performance is thus perpetually bounded by the initial quality of the pseudo-labels. Third, the quality of pseudo-labels and the detector's performance are intrinsically interdependent—a better detector yields more accurate pseudo-labels, and more accurate pseudo-labels in turn train a stronger detector. Such a virtuous cycle can only be established through alternating iterations of the E-step and M-step; a single warm-up stage alone cannot form this positive feedback mechanism. To verify whether prolonging warm-up could compensate for this deficiency, we further extended the warm-up period from 200 to 500 epochs. Experimental results show that the NAUC on the Gulfport dataset only improved from 0.355 to 0.382, still far below the 0.501 achieved by the full EM model. This confirms that merely extending warm-up cannot replace the iterative refinement mechanism, because the initial detector errors are propagated and amplified rather than corrected through re-calibration in the E-step. This result also indirectly demonstrates the model's potential to converge to a superior solution.

Variant 5 results in a 2.1% decrease in AUC on the simulated dataset and a 5.4% drop in NAUC on the real-world dataset, also causing significant performance degradation. This finding suggests that global contextual information is indispensable for enhancing feature representation. Particularly in real-world data, local receptive fields are insufficient to capture the semantic relationships between targets and their surrounding environment, whereas global information helps the model establish a more comprehensive scene understanding, thereby enabling accurate target localization in complex backgrounds. The results of Variant 6 demonstrate that the dynamic graph structure captures nonlinear spatial dependencies among pixels more effectively than fixed convolutional kernels. The 1-D convolution employs fixed weights and a fixed receptive field, making it difficult to adapt to the nonstationary distribution of spatial neighborhoods in hyperspectral images. In contrast, LS-GCM dynamically constructs the adjacency matrix based on spectral similarity, enabling adaptive aggregation of neighborhood information according to the spectral affinities among pixels. The more pronounced performance degradation on real-world scenes indicates that the adaptability advantage of the dynamic graph structure becomes increasingly prominent in complex environments. Variant 7 shows that the multiscale positional encoding based on spatial statistics outperforms conventional frequency-based encoding methods. Sinusoidal encoding relies solely on the absolute index of pixels in a 1-D

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:51:51 UTC from IEEE Xplore. Restrictions apply.

sequence and fails to reflect the 2-D spatial structure of hyperspectral images. By contrast, MCPE provides rich multiscale spatial context for each pixel through grid partitioning and sliding-window statistics. The more significant degradation on real data suggests that in scenes with complex spatial layouts, the structured positional priors of MCPE play a more critical role in compensating for model inference.

In summary, the performance degradation of all modules on the real datasets is generally more pronounced than that on the simulated datasets, indicating that the simulated data are based on idealized assumptions, where the contributions of individual modules are more easily substituted by redundant information. In contrast, real hyperspectral images involve complex land-cover distributions, illumination variations, mixed pixels, and other interferences, each of which poses practical challenges that each designed module must address, thereby further amplifying its importance in complex environments. The ablation study comprehensively verifies the effectiveness and necessity of each module in GSPT-MIL, providing a solid theoretical foundation for the superior performance of the proposed model.

E. Parameter Tuning

To investigate the influence of different hyperparameters on the detection performance of GSPT-MIL and to select the optimal configuration for the model, this section presents a detailed analysis of several key parameters. The rationale for setting the remaining parameters is provided based on empirical experience or References to the literature. Experiments are carried out on the simulated dataset with an average target abundance ratio of 0.3 and the brown target detection dataset from MUUFL Gulfport. A controlled variable approach is adopted, where only one parameter is varied at a time while all other parameters are kept at their default values.

The dynamic threshold coefficient λ controls the strictness of pseudo-label generation in the E-step, as defined in (22). Its value ranges between $[0, 1]$. When λ is close to 0, the threshold approaches the minimum confidence score within a bag, resulting in the majority of pixels being labeled as positive samples. Conversely, when λ is close to 1, the threshold approaches the maximum confidence score, retaining only a few high-confidence pixels. The choice of λ directly affects the quality of pseudo-labels, thereby influencing the training effectiveness of the M-step detector. A value that is too small introduces excessive noise, while a value that is too large leads to missed detections; therefore, an optimal balance must be identified. We varied λ from 0 to 1 in increments of 0.1 and observed the resulting AUC and NAUC values on the validation set, repeating each experiment five times and averaging the results. Fig. 11 illustrates the impact of λ on performance. It can be observed that on both datasets, performance remains relatively stable when λ lies within the interval $[0.2, 0.4]$, indicating that the model exhibits a certain degree of robustness to this parameter. When $\lambda < 0.2$, the excessive number of noisy samples leads to a sharp decline in performance; when $\lambda > 0.6$, the issue of missed detections becomes severe, also causing performance degradation. Accordingly, we select $\lambda = 0.25$ as the default value.

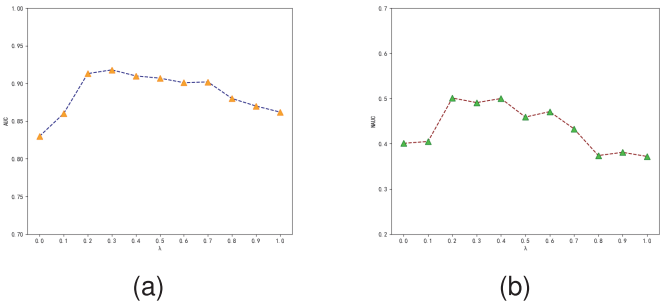


Fig. 11. Variation of the model's average AUC/NAUC with respect to the coefficient λ. (a) Simulated dataset ($\bar{P}_t = 0.3$). (b) Brown target on the MUUFL Gulfport dataset.

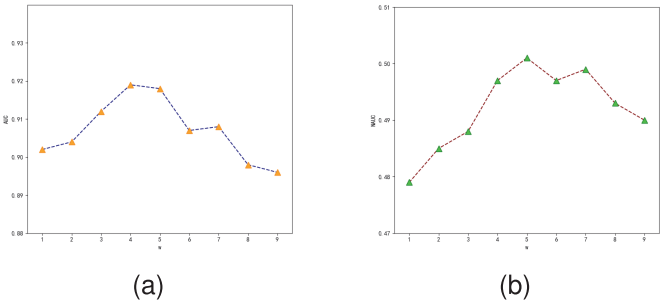


Fig. 12. Variation of the model's average AUC/NAUC with respect to the coefficient w. (a) Simulated dataset ($\bar{P}_t = 0.3$). (b) Brown target on the MUUFL Gulfport dataset.

The neighborhood window radius w in LS-GCM controls the spatial range of each pixel during graph message passing, thereby determining the scale of spatial contextual information that the model can aggregate. A too small w fails to provide sufficient neighborhood support for the target pixel, making it difficult to effectively enhance target responses, especially in subpixel target scenarios; conversely, a too large w may introduce an excessive number of heterogeneous background pixels, leading to over-smoothing of the image and weakening the separability between targets and background. Therefore, w must strike a balance between adequately capturing local context and suppressing noise interference. To determine the optimal w, we conducted sensitivity experiments on two datasets, varying w from 1 to 9, with results shown in Fig. 12. It can be observed that both datasets achieve near-optimal performance around $w = 4$ or 5, and the performance fluctuation remains within 1 percentage point for $w \in [3, 6]$, indicating that the model is reasonably robust to this parameter. Considering these findings comprehensively, we adopt $w = 5$ as the default value in this article.

The contrastive loss weight α controls the relative contribution of the classification loss and the contrastive alignment loss in the M-step, determining the strength of the constraint imposed by the SS-FAN module on feature learning. A too small α weakens the feature alignment effect, failing to fully exploit the benefits of cross-modal contrastive learning; conversely, a too large α may interfere with the primary classification task, causing the detector optimization to deviate from the correct direction. Therefore, α must strike a balance between feature alignment and the main task. To determine

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:51:51 UTC from IEEE Xplore. Restrictions apply.

TABLE V
SENSITIVITY ANALYSIS OF THE CONTRASTIVE LOSS WEIGHT α

α	Simulated Dataset ($\bar{p}_t = 0.3$) AUC	MUUFL Gulfport (brown) NAUC
0	0.915	0.497
0.05	0.918	0.499
0.1	0.918	0.501
0.2	0.917	0.500
0.5	0.911	0.491

the optimal α, we conducted sensitivity experiments on two datasets, with results shown in Table V. The results indicate that the performance is lowest when $\alpha = 0$, verifying the effectiveness of the SS-FAN feature alignment mechanism. The performance remains stable for $\alpha \in [0.05, 0.2]$, suggesting that the model is insensitive to α within this range. When $\alpha = 0.5$, the contrastive loss weight becomes excessively large, disrupting the primary classification task and leading to a notable performance drop. Taking these factors into consideration, we adopt $\alpha = 0.1$ as the default value in this article.

In addition to the aforementioned key parameters, GSPT-MIL involves several other parameters that have a relatively minor impact on performance or can be readily determined based on dataset characteristics. The rationale for setting these parameters is briefly explained below. The grid size P and sliding window size S are two parameters of the MCPE module that control the granularity of position encoding and the range of contextual information, respectively. Preliminary experiments indicate that $P = 5$ and $S = 7$ yield stable performance across most datasets, aligning well with the spatial resolution of the imagery and effectively covering the regions where targets are likely to appear. The feature dimension d_h is set to 256, a common choice that balances model capacity and computational efficiency; higher dimensions increase the risk of overfitting, whereas lower dimensions may inadequately represent spectral information. The batch size B is set to 128, maximizing training efficiency within the memory constraints while ensuring stability in gradient estimation. The number of warm-up epochs W is set to 200, based on the observation that the model acquires initial discriminative capability by this point, providing a stable starting point for subsequent EM alternation. The above parameters demonstrate consistent performance across both simulated and real-world datasets, indicating that GSPT-MIL exhibits strong robustness to parameter selection.

F. Analysis of Time Cost

To evaluate the computational efficiency of GSPT-MIL, we recorded the training and testing times of all compared methods on the simulated datasets, with results presented in Table VI. It should be noted that different methods were implemented on different software platforms and hardware configurations; therefore, the absolute time comparisons are provided only for reference, while the relative differences still offer valuable insights.

TABLE VI
COMPARISON OF EXECUTION TIME ON SIMULATED DATASETS

Alg	Train(s)	Test(s)	Platform
EM-DD	45.958	0.738	MATLAB®2024a Intel®Core™i5-1135G7 @2.40 GHz CPU
MILES	124.515	5.282	
MI-HE	669.240	7.812	
MI-ACE	154.926	0.731	
MI-SMF	164.545	0.728	
MIMD-ACE	23.008	0.762	
MIMD-SMF	25.495	5.334	
Attention-DMIL	4.752	2.957	Python: 3.9.7 PyTorch: 2.8.0 (CPU version)
VMIL-ECML	7.263	4.364	
GSPT-MIL	14.031	6.297	

As can be seen from Table VI, traditional concept-learning-based methods generally require longer training times. In particular, MI-HE needs to iteratively optimize target and background dictionaries, taking over 660 s, whereas the MIMD series, benefiting from a more efficient initialization strategy, substantially reduces the training time to approximately 25 s. During the testing phase, traditional methods generally exhibit faster inference speeds, as their detectors rely on closed-form solutions or simple matching operations.

For deep learning methods, the reported training time in the table refers to the average time per epoch. GSPT-MIL takes 14.031 s per epoch, approximately 1.93 times that of VMIL-ECM, mainly due to two additional overheads. The first is the spatial information modeling cost: the graph construction and multiple rounds of message passing in LS-GCM, along with multiscale sliding-window statistics and positional encoding fusion in MCPE, together constitute the primary computational burden. The second is the feature alignment cost, where the contrastive loss computation in the SS-FAN module requires projection and within-batch pairwise similarity calculations, adding extra matrix operations. Nevertheless, the per-epoch time of GSPT-MIL remains significantly lower than the per-iteration time of traditional methods, and can be further reduced through GPU parallel acceleration. In the testing phase, GSPT-MIL takes 6.297 s, which is higher than VMIL-ECM and Attention-DMIL, but all deep learning methods are significantly faster than traditional ones. The main overhead in the testing phase of GSPT-MIL arises from the positional sequence generation in MCPE and the graph-structure inference in LS-GCM; however, since backpropagation and EM iterations are not required during inference, this time remains within an acceptable range and is sufficient for practical offline batch-processing scenarios.

In summary, although GSPT-MIL incurs a higher per-epoch training time compared to other deep learning methods, it still exhibits a significant efficiency advantage over traditional methods; the testing time overhead is within a reasonable range, demonstrating its potential for practical applications. Future work may explore strategies such as structural pruning of graph convolution layers and lightweight design of positional encodings to further reduce the computational cost of spatial information modeling.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:51:51 UTC from IEEE Xplore. Restrictions apply.

G. Discussion of Limitations

Despite the superior detection performance achieved by GSPT-MIL on multiple datasets, its effectiveness is still limited under certain data conditions. First, the proposed method assumes that hyperspectral images exhibit clear spatial continuity, i.e., target pixels are distributed in a clustered manner in the spatial domain, and the graph-structure modeling of LS-GCM leverages this characteristic to aggregate neighborhood information. However, when targets are present in an extremely sparse or randomly scattered manner, the amount of valid information that can be aggregated from local neighborhoods decreases substantially, and the benefits of graph message passing may consequently diminish. Second, the method relies on the quality of bag-level annotations; if the GPS positioning error is excessively large or the positively labeled regions fail to fully cover the targets, the pseudo-label initialization in the E-step will be adversely affected, thereby interfering with the convergence direction of the subsequent EM iterations.

In terms of computational efficiency, the per-epoch training time of GSPT-MIL is approximately 1.93 times that of VMIL-ECM, primarily attributed to the graph construction and message passing in LS-GCM, the multiscale sliding-window statistics in MCPE, and the contrastive projection computations in SS-FAN. The current implementation is suitable for offline training scenarios. For applications with stringent real-time requirements, such as airborne or spaceborne platforms, further reduction of inference time is still needed through strategies including model pruning, graph sparsification, or precomputation of positional encodings. Nevertheless, in the trade-off between detection accuracy and computational cost, GSPT-MIL still demonstrates strong practical value in most nonreal-time application scenarios.

IV. Conclusion

This article addresses the challenges of strong prior knowledge dependence and the difficulty of obtaining pixel-level annotations in weakly supervised hyperspectral object detection by proposing a graph structure-prior embedded multi-instance learning GSPT-MIL method based on the EM framework. This method innovatively integrates an LS-GCM, a MCPE, and an SS-FAN into the EM optimization process, enabling the collaborative inference of object locations and the learning of discriminative features using only region-level labels. Through systematic validation on both simulated and real-scene datasets, GSPT-MIL significantly outperforms existing weakly supervised and traditional methods across multiple evaluation metrics, demonstrating its effectiveness and superiority in achieving accurate pixel-level detection without relying on target spectral priors. This research not only provides a novel and powerful solution for weakly supervised hyperspectral analysis but also offers valuable insights for related fields exploring structurally aware and cross-modal aligned deep learning models. Future work will focus on further optimizing algorithm efficiency and exploring the model's generalization capability in a wider range of remote sensing scenarios.

References

[1] P. Liu, Y. Bu, Y. Zhao, and S. G. Kong, "Progressive self-supervised framework for anomaly detection in hyperspectral images," *Eng. Appl. Artif. Intell.*, vol. 156, Sep. 2025, Art. no. 111151, doi: 10.1016/j.engappai.2025.111151.

[2] Y. Li, Y. Luo, L. Zhang, Z. Wang, and B. Du, "MambaHSI: Spatial–spectral mamba for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5524216, doi: 10.1109/TGRS.2024.3430985.

[3] Y. Li, Y. Shi, K. Wang, B. Xi, J. Li, and P. Gamba, "Target detection with unconstrained linear mixture model and hierarchical denoising autoencoder in hyperspectral imagery," *IEEE Trans. Image Process.*, vol. 31, pp. 1418–1432, 2022, doi: 10.1109/TIP.2022.3141843.

[4] C. Kishor Kumar Reddy et al., "Enhancing precision agriculture and land cover classification: A self-attention 3D convolutional neural network approach for hyperspectral image analysis," *IEEE Access*, vol. 12, pp. 125592–125608, 2024, doi: 10.1109/ACCESS.2024.3420089.

[5] M. Shen et al., "Challenges in remote sensing of vegetation phenology," *Innov. Geosci.*, vol. 2, Jan. 2024, Art. no. 100070.

[6] Q. Li, J. Li, T. Li, and Y. Feng, "A joint framework for underwater hyperspectral image restoration and target detection with conditional diffusion model," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 17263–17277, 2024, doi: 10.1109/JSTARS.2024.3464557.

[7] Y. Xu, B. Du, and L. Zhang, "Robust self-ensembling network for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 3, pp. 3780–3793, Mar. 2024, doi: 10.1109/TNNLS.2022.3198142.

[8] A. E. Adetutu, Y. F. Bayo, A. A. Emmanuel, and A. A. A. Opeyemi, "A review of hyperspectral imaging analysis techniques for onset crop disease detection, identification and classification," *J. Forest Environ. Sci.*, vol. 40, no. 1, p. 80, 2024.

[9] J. Gong, Z. Huang, Z. Yang, X. Ding, and F. Li, "Spectral information divergence-driven diffusion networks for hyperspectral target detection," *Appl. Sci.*, vol. 15, no. 8, Apr. 2025, Art. no. 4076, doi: 10.3390/app15084076.

[10] C. Zhang et al., "Self-paced probabilistic collaborative representation for anomaly detection of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5516510, doi: 10.1109/TGRS.2024.3393303.

[11] M. Wang, B. Yang, H. Zhang, C. Wang, X. Jiang, and C. Jiao, "Hyperspectral target detection via multi-instance self-attention semantic feature extraction," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2024, pp. 9450–9454, doi: 10.1109/IGARSS53475.2024.10641595.

[12] T. G. Dietterich, R. H. Lathrop, and T. Lozano-Pérez, "Solving the multiple instance problem with axis-parallel rectangles," *Artif. Intell.*, vol. 89, nos. 1–2, pp. 31–71, Jan. 1997.

[13] M.-A. Carbonneau, V. Cheplygina, E. Granger, and G. Gagnon, "Multiple instance learning: A survey of problem characteristics and applications," *Pattern Recognit.*, vol. 77, pp. 329–353, May 2018, doi: 10.1016/j.patcog.2017.10.009.

[14] J. Bolton and P. Gader, "Application of multiple-instance learning for hyperspectral image analysis," *IEEE Geosci. Remote Sens. Lett.*, vol. 8, no. 5, pp. 889–893, Sep. 2011, doi: 10.1109/LGRS.2011.2135330.

[15] A. Zare and C. Jiao, "Extended functions of multiple instances for target characterization," in *Proc. 6th Workshop Hyperspectral Image Signal Processing: Evol. Remote Sens. (WHISPERS)*, Jun. 2014, pp. 1–4.

[16] C. Jiao and A. Zare, "Functions of multiple instances for learning target signatures," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 8, pp. 4670–4686, Aug. 2015, doi: 10.1109/TGRS.2015.2406334.

[17] A. Zare, C. Jiao, and T. Glenn, "Discriminative multiple instance hyperspectral target characterization," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 10, pp. 2342–2354, Oct. 2018, doi: 10.1109/TPAMI.2017.2756632.

[18] C. Jiao, B. Yang, Q. Wang, G. Wang, and J. Wu, "Discriminative multiple-instance hyperspectral subpixel target characterization," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5521420, doi: 10.1109/TGRS.2022.3140798.

[19] C. Jiao, B. Yang, and J. Wu, "Multiple instance constrained energy minimization for discriminative hyperspectral target characterization," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2021, pp. 3197–3200.

[20] C. Jiao and A. Zare, "Multiple instance hybrid estimator for learning target signatures," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 988–991.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:51:51 UTC from IEEE Xplore. Restrictions apply.

[21] C. Jiao, C. Chen, R. G. McGarvey, S. Bohlman, L. Jiao, and A. Zare, "Multiple instance hybrid estimator for hyperspectral target characterization and sub-pixel target detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 146, pp. 235–250, Dec. 2018, doi: 10.1016/j.isprsjprs.2018.08.012.
[22] P. Zhong, Z. Gong, and J. Shan, "Multiple instance learning for multiple diverse hyperspectral target characterizations," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 31, no. 1, pp. 246–258, Jan. 2020, doi: 10.1109/TNNLS.2019.2900465.
[23] S. Meerdink et al., "Multitarget multiple-instance learning for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5502814, doi: 10.1109/TGRS.2021.3060966.
[24] X. Wang, Y. Yan, P. Tang, X. Bai, and W. Liu, "Revisiting multiple instance neural networks," *Pattern Recognit.*, vol. 74, pp. 15–24, Feb. 2018, doi: 10.1016/j.patcog.2017.08.026.
[25] Y. Chen, J. Bi, and J. Z. Wang, "MILES: Multiple-instance learning via embedded instance selection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 28, no. 12, pp. 1931–1947, Dec. 2006, doi: 10.1109/TPAMI.2006.248.
[26] M. Ilse, J. M. Tomczak, and M. Welling, "Attention-based deep multiple instance learning," in *Proc. 35th Int. Conf. Mach. Learn. (ICML)*, vol. 80, 2018, pp. 2122–2132.
[27] X. Wang et al., "Hyperspectral target detection via deep multiple instance self-attention neural network," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 2284–2287, doi: 10.1109/IGARSS.2019.8898420.
[28] C. Jiao et al., "L_1 sparsity-regularized attention multiple-instance network for hyperspectral target detection," *IEEE Trans. Cybern.*, vol. 53, no. 1, pp. 124–137, Jan. 2023, doi: 10.1109/TCYB.2021.3087662.
[29] L. Cai, S. Huang, Y. Zhang, J. Lu, and Y. Zhang, "AttriMIL: Revisiting attention-based multiple instance learning for whole-slide pathological image classification from a perspective of instance attributes," *Med. Image Anal.*, vol. 103, Jul. 2025, Art. no. 103631, doi: 10.1016/j.media.2025.103631.
[30] W. Ping, Y. Xu, K. Ren, C. H. Chi, and F. Shen et al., "Non-IID multi-instance dimensionality reduction by learning a maximum bag margin subspace," in *Proc. 24th AAAI Conf. Artif. Intell. (AAAI)*, 2010, pp. 551–556.
[31] B. Chen, L. Liu, Z. Zou, and Z. Shi, "Target detection in hyperspectral remote sensing image: Current status and challenges," *Remote Sens.*, vol. 15, no. 13, Jun. 2023, Art. no. 3223, doi: 10.3390/rs15133223.
[32] B. Yang, C. Jiao, J. Wu, and L. Li, "Variational multiple-instance learning with embedding correlation modeling for hyperspectral target detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 8, pp. 13953–13967, Aug. 2025, doi: 10.1109/TNNLS.2024.3510385.
[33] Q. Zhang and S. Goldman, "EM-DD: An improved multiple-instance learning technique," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2001, pp. 1073–1080.
[34] P. Gader, A. Zare, R. Close, J. Aitken, and G. Tuell, "MUUFL Gulfport hyperspectral and LiDAR airborne data set," Mach. Learn. Sens. Lab, Elect. Comput. Eng., Univ. Florida, Gainesville, FL, USA, Tech. Rep. REP-2013-570, 2013.
[35] T. Glenn, A. Zare, P. Gader, and D. Dranishnikov, "Bullwinkle: Scoring code for sub-pixel targets (version 1.0)," Mach. Learn. Sens. Lab, Elect. Comput. Eng., Univ. Florida, Gainesville, FL, USA, Tech. Rep., 2013. [Online]. Available: https://github.com/GatorSense/MUUFLGulfport/



Ying Cui was born in Harbin, Heilongjiang, China, in 1979. She received the Ph.D. degree in signal and information processing and the Ph.D. degree in computer science and technology from Harbin Engineering University (HEU), Harbin, in 2006 and 2009, respectively.

She joined the faculty in 2005 and is currently an Associate Professor with the School of Information and Communication Engineering, HEU. Her research interests include intelligent signal processing, image processing, and wireless sensor network optimization.

Dr. Cui is a reviewer for many journals, including *Journal of Wuhan University*, *Journal of Harbin Engineering University*, *Journal of Applied Remote Sensing*, *Acta Optica Sinica*, and *Laser and Optoelectronics Progress*.



Xiaoqi Ren received the B.E. degree in electronic information engineering from Xi'an Petroleum University, Xi'an, China, in 2024, where she is currently pursuing the M.S. degree in information and communication engineering with Harbin Engineering University, Harbin, Heilongjiang, China.

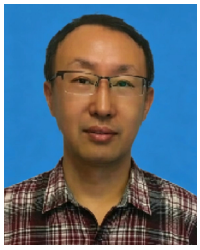
Her research interests include deep learning, computer vision, and remote sensing image processing.



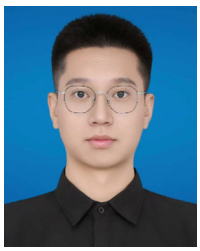
Liguo Wang (Member, IEEE) received the M.S. degree in natural science and the Ph.D. degree in engineering from Harbin Institute of Technology (HIT), Harbin, China, in 2002 and 2006, respectively, and the Ph.D. degree from Harbin Engineering University, Harbin, in 2009.

He is currently a Professor at the School of Information and Communication Engineering, Dalian Minzu University, Dalian, China.

Dr. Wang is currently a Senior Member of the Chinese Institute of Electronics and the China Institute of Communications (CIC) and a member of the Chinese Society for Optical Engineering. He is also a fellow of the International Society for Digital Earth (ISDE). He is also a member of the Chinese Society of Optical Engineering and a Senior Committee Member of the Photoelectric Technology of Astronautics Society. He is a reviewer for many journals, including IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE GEOSCIENCE AND REMOTE SENSING LETTERS, and *Journal of Electronics and Information Technology*.



Haifeng Zhu has long been engaged in research on artificial intelligence and its applications. As the technical lead, he undertook a provincial-level project on AI-based teaching application systems. As a Key Member, he participated in four projects funded by the National Natural Science Foundation of China. He has published more than ten SCI/EI-indexed academic papers as first or corresponding author, has been granted or has applied for more than ten national invention patents, and is currently revising a national-level planned textbook for the 14th Five-Year Plan.



Zhongwei Shen (Member, IEEE) received the B.Eng. degree from Xidian University, Xi'an, China, in 2018, and the Ph.D. degree from Dalian University of Technology, Dalian, China, in 2024.

Since September 2024, he has been a Lecturer with the College of Information and Communication Engineering, Harbin Engineering University, Harbin, China. His research interests include topology control and data collection protocols for underwater sensor networks.



Chunhui Zhao received the B.S. and M.S. degrees from Harbin Engineering University, Harbin, China, in 1986 and 1989, respectively, and the Ph.D. degree from the Department of Automatic Measurement and Control, Harbin Institute of Technology, Harbin, in 1998.

He was a Post-Doctoral Research Fellow with the College of Underwater Acoustical Engineering, Harbin Engineering University, where he is currently working with the College of Information and Communication Engineering as a Professor and a Ph.D. Supervisor. His research interests include digital signal and image processing, mathematical morphology, and hyperspectral remote sensing image processing.

Dr. Zhao is a Senior Member of Chinese Electronics Academy.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 13:51:51 UTC from IEEE Xplore. Restrictions apply.